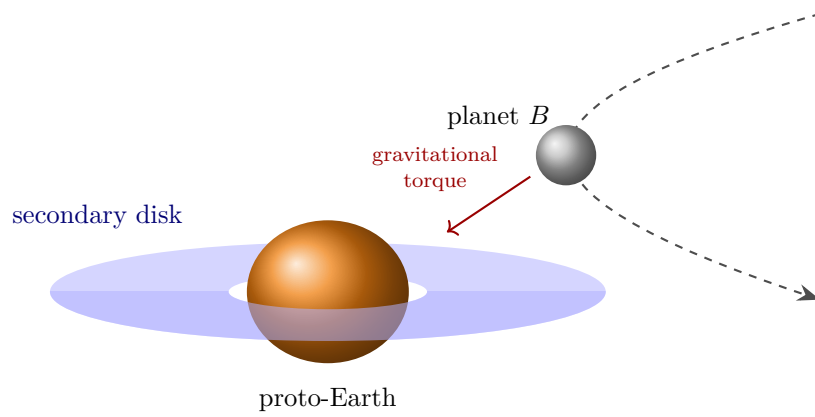


Moon Formation

Secondary Circumterrestrial Disk Theory

A retrodictive path theory
from the proto-terrestrial system to the Earth–Moon system

Reservoirs, fluxes, non-collisional planetary encounter,
geochemistry and viability domain



Michel Debailleul

Geophysicist — M.Sc., Université libre de Bruxelles (ULB)

Independent researcher

ORCID: 0009-0003-1222-1433 | michel.debailleul@yahoo.fr

© 2026 Michel Debailleul — License: CC BY-NC 4.0

Abstract

The Earth–Moon system currently exhibits a dynamical architecture, a mass distribution, an angular momentum, isotopic signatures, geochemical compositions and a chronology that together form independent observational constraints. The question addressed here is not to describe these properties, but to identify a continuous *physical path* linking the initial state of the proto-terrestrial system to the observed state.

The astrophysical anchor of this approach is that natural satellites are already interpreted as the products of circumplanetary reservoirs: the regular satellites of Jupiter and Saturn in circumplanetary disks, and the inner satellites of Neptune in a reworked system where a debris disk probably followed the capture of Triton. The proposed terrestrial case does not import their masses or compositions; it retains the physical principle of a reservoir bound to a planet, able to transport mass and angular momentum and then to form satellites.

This work explores such a path. A secondary circumterrestrial disk, contemporaneous with the accretion of the proto-Earth, forms an orbital reservoir of mass and angular momentum before the Moon forms. A planet B , on a grazing and non-collisional passage, does not supply the bulk of the lunar material: it acts as a *transient gravitational agent redistributing mass and angular-momentum fluxes* between four reservoirs — the proto-Earth, the disk, planet B and the external medium.

The heliocentric logic of the model is that B need not survive in the present Solar System. It may belong to the population of primordial planetary bodies later eliminated by the global orbital instability of the Solar System. This extension does not necessarily identify it with a fifth ice giant; it only requires its grazing passage to precede its dynamical elimination, according to $t_B < t_{ej}$. It thus links the Earth–Moon problem to the broader problem of primordial planetary migration and instability [53, 54, 55, 56, 57, 58].

The passage performs an orbital sorting. A bound cohort of matter acquires an angular momentum that places it, after circularization, beyond the Roche limit, in the window where the Moon can accrete; one fraction re-accretes onto the proto-Earth, another escapes. A passage parallel to the dynamical low-latitude band of an oblate rapidly rotating proto-Earth can also inject into the disk surface material already depleted in metal and siderophile elements; W, V, Cr, Mn, FeO, O and Ti then provide tracers with which to test the lineage.

The work successively formulates the scientific problem, the astrophysical foundations, the reservoir architecture, the physical chronology, the passage dynamics, the evolution of the circumterrestrial reservoir, geochemistry by tracer families, the viability domain and the predictions, each with a verification method. It does not claim to prove a unique history: it defines a quantitative and falsifiable physical path.

Keywords: Moon formation; circumterrestrial disk; secondary accretion; non-collisional encounter; orbital sorting; angular momentum; Roche limit; lunar geochemistry; primordial orbital instability; eliminated planet.

Methodological Preamble

A Retrodictive Path Theory

This monograph does not claim to reconstruct uniquely what actually happened in the history of the Solar System. Such a reconstruction is probably inaccessible: the exact initial conditions of the proto-terrestrial system, the individual trajectories of planetary bodies, the detailed thermodynamic states of the reservoirs, and the fine chronology of encounters have disappeared.

The aim is different. The purpose is to construct a *possible physical path* linking an admissible proto-terrestrial system to the observed Earth–Moon system. The approach is therefore retrodictive: it starts from the known final state and moves backward, constraint after constraint, toward a family of initial states and dynamical steps capable of producing that final state.

The problem addressed here is therefore not:

Which unique event actually occurred?

but rather:

Does there exist a continuous, quantitatively constrained and falsifiable physical path linking a possible proto-terrestrial state to the observed Earth–Moon state?

This status may be written as

$$\boxed{\exists \Gamma(t) \in \mathcal{D}_{\text{viable}} \quad \text{such that} \quad \Gamma(t_0) = \mathcal{S}_0, \quad \Gamma(t_f) = \mathcal{S}_f.} \quad (1)$$

The trajectory $\Gamma(t)$ does not necessarily denote the real history of the Solar System. It denotes an admissible path in the space of physical states: mass, angular momentum, energy, composition, phases of matter, and chronology. The scientific value of the theory therefore does not rest on a claim of historical uniqueness, but on the possible existence of a continuous domain of parameters satisfying, simultaneously, the dynamical, geochemical, isotopic, thermodynamic and chronological constraints.

In this framework, the secondary circumterrestrial disk is not introduced as a narrative device, but as a physical reservoir to be tested. Likewise, planet B does not supply the bulk of the lunar material: it is a transient gravitational agent redistributing mass and angular-momentum fluxes. Its subsequent fate belongs to the same retrodictive framework. It may belong to the population of planetary bodies of the primitive Solar System later removed by collision, accretion, gravitational scattering or ejection.

The minimal condition associated with this heliocentric extension is

$$\boxed{B \in \mathcal{P}_{\text{elim}}, \quad t_B < t_{\text{ej}}.} \quad (2)$$

Here, $\mathcal{P}_{\text{elim}}$ denotes the population of planetary bodies present in the primitive Solar System but absent from the present one; t_B is the time of the grazing passage near the proto-Earth; and t_{ej} is the time of the dynamical removal of B . This formulation leaves open both possibilities: an early orbital instability or a later instability.

Status of the theory

The theory of the secondary circumterrestrial disk is a path theory, not a claim of historical uniqueness. It becomes credible if a continuous domain of parameters allows the dynamical, geochemical, isotopic, thermodynamic and chronological gates to be crossed together. It is falsified if no such domain exists.

Reading guide

The work is built to be read, not merely consulted. Each part begins with the question it addresses and ends by linking to the next one. Long demonstrations are collected in Appendix A, so that the main text remains continuous; notation is collected in Appendix B, and numerical values in Appendix C.

Distinct kinds of boxes guide the reading:

Definition

Defines the meaning of a term or a role, which then becomes the reference for the whole work.

Physical idea

Gives the physical intuition behind a calculation.

Physical reading

Interprets an equation or result within the physical path.

Didactic reminder

Recalls a classical notion needed for understanding.

Quantitative control

Compares a result with orders of magnitude or with a conservation balance.

Observable consequence

States what should be observed, in lunar samples or simulations, if the described mechanism is at work.

Falsification criterion

States the condition under which a branch of the theory must be abandoned.

Original contribution of the theory

Highlights what the proposed path contributes in its own right.

Contents

Methodological preamble	ii
I The scientific problem	1
1 From the proto-terrestrial system to the Earth–Moon system	2
1.1 The scientific problem	2
1.2 Observational constraints	2
1.3 Astrophysical anchor: natural satellites from disks	2
1.4 Hypothesis explored	3
1.5 Heliocentric fate of B and eliminated planet	4
1.6 Scope of the theory	5
1.7 Status of the elements of the path	5
1.8 Organization of the work	6
II Astrophysical foundations	7
2 The principle of secondary accretion	8
2.1 The circumplanetary principle	8
2.2 Physical meaning of the term “secondary”	8
2.3 Three nested propositions	8
2.4 A hierarchy of accretion systems	9
2.5 Jupiter: the Galilean system as a dynamical reference	9
2.6 Saturn: viscous transport, Roche limit and satellite birth	10
2.7 Uranus: a regular system with a still-debated reservoir origin	10
2.8 Neptune: Triton’s capture, debris disk and reaccretion	10
2.9 Direct observation: the circumplanetary disk of PDS 70c	10
3 From the principle to the terrestrial case	12
3.1 Quantitative comparison of natural systems	12
3.2 From several satellites to one dominant satellite	13
3.3 Specificity of the terrestrial case: a solid-dominated disk	13
III Physical architecture: the four reservoirs	15
4 Reservoirs, fluxes and conservation laws	16
4.1 The four reservoirs of the system	16
4.2 Mass fluxes	17
4.3 Angular momentum	17
4.4 Energy	17
4.5 Composition	18
4.6 Level I state variables	18
4.7 Encounter parameters	18
4.8 Reference scales of the proto-Earth–disk– B system	19

IV	Physical chronology	20
5	Physical chronology of the path	21
5.1	Phases defined by dominant fluxes	21
5.2	Hierarchy of time scales	21
5.3	Order of stages	22
5.4	Gaseous disk and residual reservoir	22
5.5	Three chronological regimes	22
V	Dynamics: the gravitational agent B	24
6	The non-collisional passage	25
6.1	A gravitational interaction without contact	25
6.2	Geometrical definition of the non-collisional character	25
6.3	Dynamical regime of the encounter	26
6.4	Exact differential field	26
6.5	Torque exerted on the disk	26
6.6	Low-latitude geometry of the passage	26
6.7	Angular-momentum reservoir carried by B	28
6.8	Integrity of B during the encounter	29
6.9	After the passage: return to the heliocentric problem	29
7	Injection of superficial proto-terrestrial matter	30
7.1	Role of Earth's rotation	30
7.2	Injection threshold	30
7.3	Transient potential	30
7.4	Injected matter flux	31
8	The lunar window: the passage as orbital sorting	32
8.1	Criterion for membership in the lunar window	32
8.2	A geometrical relation between the three conditions	32
8.3	Measuring the success of the passage	32
8.4	Numerical objective	33
8.5	Anatomy of sorting: the tangential impulse	33
8.6	Tidal impulse: analytical form and order-of-magnitude bound	34
8.7	From orbital sorting to a dominant satellite	35
9	Balances of the encounter	36
9.1	Mass balance	36
9.2	Angular-momentum balance	36
VI	The secondary disk as an evolving reservoir	37
10	Feeding and structure of the reservoir before the passage	38
10.1	Accretional origin of the reservoir	38
10.1.1	Matter supply in the Hill sphere	38
10.1.2	Mass and surface density	38
10.1.3	Angular momentum supplied by accretion	38
10.1.4	Origin of proto-terrestrial spin	39
10.2	Radial structure	39
10.2.1	Inner radius	39

10.2.2	Outer radius	39
10.2.3	Roche limit and synchronous orbit	39
10.3	The reservoir as an accretion disk	40
10.4	Orbital structure and angular-momentum transport	41
10.4.1	Quasi-Keplerian rotation	41
10.4.2	Global disk angular momentum	41
10.4.3	Viscous evolution	41
11	Thermodynamics, stability and persistence	42
11.1	Thermal and multiphase state	42
11.1.1	Energy balance	42
11.1.2	Condensation and phase transitions	42
11.1.3	Vertical settling	42
11.2	Gravitational stability and persistence	43
11.2.1	Toomre criterion	43
11.2.2	Disk persistence	43
12	Reservoir reorganization and lunar accretion	44
12.1	New state of the disk	44
12.2	Dissipation	44
12.3	Mass available beyond Roche	44
12.4	Disk–satellite accretion control	44
VII	Geochemistry: families of tracers	46
13	Three material lineages	47
13.1	Fractions of provenance	47
13.2	Concentration-weighted isotopic mixing	47
13.3	One physical question per tracer family	47
14	Families of tracers	48
14.1	Moderately siderophile refractory elements: V, Cr, Mn	48
14.2	Tungsten: abundance, isotopes and clock	48
14.3	Iron and siderophiles: FeO, Ni, Co and metal	49
14.3.1	FeO and silicate reservoir	49
14.3.2	Metal and silicate	50
14.4	O, Ti, Cr isotopes: the imprint of B	50
14.5	Volatiles	50
14.6	The W/V ratio and depth of origin	50
14.7	Synthesis of tracers	51
VIII	Viability domain	52
15	Parameters, gates and viable domain	53
15.1	Level I parameters	53
15.2	Gates of the viable domain	53
16	Level I numerical program	56
16.1	Phase A: disk formation and evolution before B	56
16.2	Phase B: grazing passage of B	56
16.3	Phase C: relaxation and accretion	56

16.4 Phase B protocol and viability maps	56
16.5 Injection budget: evaluating $M_{\oplus \rightarrow D}$	57
16.6 First pilot calculation of orbital sorting	58
IX Predictions	62
17 Predictions and verification	63
General conclusion	65
A Demonstrations	66
A.1 Circularization radius of captured matter	66
A.2 Frequency of grazing passages	66
A.3 Trajectory nodes and non-crossing	66
A.4 Circularization radius and perigee	66
A.5 Orbital-sorting boundaries	66
A.6 Tidal impulse: analytical form under hypotheses	67
A.7 Circularization energy	68
A.8 Injection threshold	68
A.9 Bound on the chondritic share of the reservoir	68
B Notation	69
C Constants and numerical values	70
Bibliography	71

Part I

The scientific problem

This first part states the question answered by the work. It begins neither with the disk nor with planet B , but with the Earth–Moon system as it is observed, and with the requirement these observations impose on any account of its formation. The secondary disk, planet B , fluxes between reservoirs, geochemistry and dynamics then appear as elements of a single answer.

Chapter 1

From the proto-terrestrial system to the Earth–Moon system

1.1 The scientific problem

The Moon exists. The Earth–Moon system currently has a dynamical architecture, a mass distribution, an angular momentum, isotopic signatures, geochemical compositions and a chronology that constitute independent observational constraints.

These observations are now numerous and precise enough to impose a demanding physical framework on any theory of lunar formation. None of these constraints can be considered in isolation: orbital dynamics, geochemistry, isotopic systems, thermodynamics and chronology describe one and the same physical system and must be explainable coherently.

The scientific question is therefore not whether the Moon exists or what its present properties are, but to determine:

*What physical path naturally led from the proto-terrestrial system
to the presently observed Earth–Moon system?*

In other words, the task is to identify a physically admissible evolution linking an initial state of the proto-Earth to the final state represented by the Earth–Moon system, while simultaneously respecting the constraints imposed by observations. This approach may be represented generally as

$$\boxed{\mathcal{S}_0 \longrightarrow \mathcal{S}_f} \quad (1.1)$$

where $\mathcal{S}_0$ denotes the initial state of the proto-terrestrial system and $\mathcal{S}_f$ the currently observed state. The scientific problem is to determine whether a continuous physical path exists between these two states.

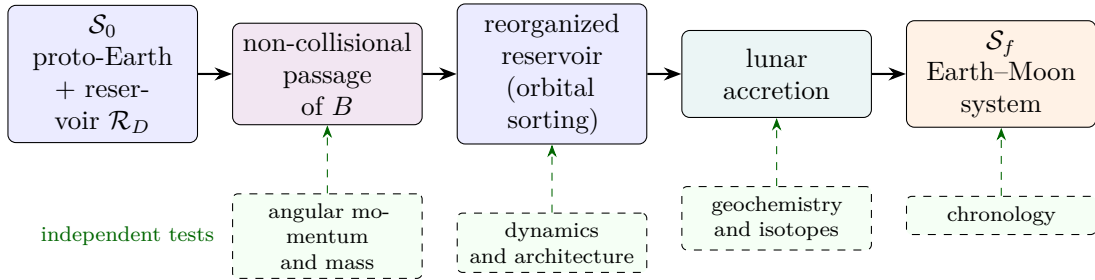


Figure 1.1: The problem formulated as the search for a continuous physical path between the initial state $\mathcal{S}_0$ and the observed state $\mathcal{S}_f$. Each family of observations tests the path independently.

1.2 Observational constraints

Table 1.1 gathers the families of constraints that the path must satisfy simultaneously. They will serve, in Part IX, as independent tests.

1.3 Astrophysical anchor: natural satellites from disks

Before introducing the terrestrial case, one astrophysical fact must be stated explicitly: the formation of satellites from a circumplanetary reservoir is not an ad hoc construction. In the Solar

Table 1.1: Families of observational constraints imposed on the physical path.

Family	Constraint	Nature or order of magnitude
Mass	ratio $M_L/M_\oplus$	1.23×10^{-2}
Architecture	one dominant satellite	$f_{\text{dom}} \simeq 1$
Angular momentum	total system angular momentum	$L_{\text{EM}} \approx 3.5 \times 10^{34} \text{ kg m}^2 \text{ s}^{-1}$
Orbit	lunar orbital inclination	about 5° relative to the ecliptic
Internal structure	small lunar core	at most a few percent of the mass
Isotopes	O, Ti, Cr, W	Earth and Moon very close, often indistinguishable
Siderophiles	V, Cr, Mn, W	depletions similar to those of Earth's mantle
Iron	FeO in the lunar mantle	higher than in Earth's mantle
Volatiles	K/U, Zn, isotopes	Moon depleted in volatile elements
Chronology	formation age	several tens of millions of years after the first solids

System, several families of natural satellites are interpreted as the products of matter bound to a planet, organized in a disk, a massive ring or a debris disk, and then transformed into satellites by accretion, migration, dissipation and crossing of the Roche limit.

Observational anchor

The theory does not start from the abstract idea that a disk could form a Moon. It starts from a principle observed more than once:

$$\boxed{\text{growing or reworked planet} + \text{circumplanetary reservoir} \longrightarrow \text{natural satellites.}} \quad (1.2)$$

The terrestrial case must then be treated with its own constraints: a terrestrial planet, a solid-dominated reservoir, a high lunar mass and the non-collisional passage of B acting as the agent of redistribution.

Three examples play a particular role in the reasoning. The Galilean system of Jupiter is the classical example of a regular satellite system whose formation is linked to a circumjovian disk fed during the growth of the planet. Models discuss gas and solid inflow, angular-momentum transport, migration and the conditions of satellite accretion [14, 46, 10, 15]. Saturn provides a second anchor: its regular satellites and ring system show that a circumplanetary reservoir can transport matter, spread viscously, cross the Roche limit and form satellites [16, 47, 17]. Neptune is subtler: Triton is very probably a captured object, but its arrival may have destroyed or reworked a prior regular system and produced a debris disk from which the present inner satellites reformed or reaccreted [48, 49, 50, 51, 52].

This distinction is essential. The satellites of Jupiter, Saturn and Neptune do not prove the history of the Moon. They show that nature already possesses the basic mechanism: storing matter in a circumplanetary reservoir, transporting angular momentum within it and converting part of that reservoir into satellites. The proposed theory applies this principle to the proto-terrestrial case and adds its own step: the grazing non-collisional passage of B , which does not create the reservoir but sorts and reorganizes it.

1.4 Hypothesis explored

The present theory explores one possible path. It applies to the proto-terrestrial case the physical principle established by natural satellite systems: a planet can possess a circumplanetary reservoir

Table 1.2: Natural examples showing that a circumplanetary reservoir can form or reform satellites. These examples establish the physical principle; they do not set the parameters of the terrestrial case.

System	Reservoir or process invoked	Physical lesson for the proposed path
Jupiter	circumjovian disk fed by gas and solids during the growth of Jupiter	a circumplanetary disk can produce several organized regular satellites
Saturn	circumplanetary disk, massive rings, viscous spreading and crossing of the Roche limit	a particulate reservoir can migrate, cross Roche and feed satellite accretion
Neptune	capture of Triton, destruction or reworking of a previous system, debris disk and reaccretion of the inner satellites	an external gravitational event can reorganize a circumplanetary system and leave a reservoir from which satellites reform

able to store matter, transport angular momentum and form satellites. It is based on the hypothesis that a secondary circumterrestrial disk, contemporaneous with the accretion of the proto-Earth, formed such an orbital reservoir of matter and angular momentum before the Moon formed.

In this framework, a planetary body B does not supply the bulk of the lunar material. Its passage remains non-collisional and acts as a transient gravitational agent that redistributes mass and angular-momentum fluxes between the different reservoirs of the proto-terrestrial system.

Definition

Role of planet B . Planet B is a *transient gravitational agent redistributing mass and angular-momentum fluxes between the reservoirs of the proto-terrestrial system*. It does not supply the bulk of the lunar material. Its action begins and ends with its passage. This definition is the reference for the entire work.

The Moon then results from the evolution of this reorganized system, mainly from matter already present in the circumterrestrial disk, possibly enriched by a limited contribution from the superficial layers of the proto-Earth.

1.5 Heliocentric fate of B and eliminated planet

Defining B as a transient agent immediately raises a legitimate question: if a planet grazed the proto-Earth without colliding with it, where is it today? The theory answers by connecting the Earth–Moon problem to the global evolution of the primordial Solar System. B need not be a body that still survives; it may belong to the population of planetary bodies eliminated during the primordial orbital instabilities.

We denote this population by

$$\mathcal{P}_{\text{elim}} = \left\{ \begin{array}{l} \text{primordial planetary bodies eliminated by collision, accretion,} \\ \text{gravitational diffusion or ejection} \end{array} \right\}. \quad (1.3)$$

The minimal heliocentric proposal is therefore

$$B \in \mathcal{P}_{\text{elim}}, \quad t_B < t_{\text{ej}} \quad (1.4)$$

where t_B is the time of the grazing passage near the proto-Earth, and t_{ej} is the time of the ejection or dynamical elimination of B .

This formulation is deliberately more robust than the strong statement “ B is the missing planet”. Models of outer Solar System instability show that migration of the giant planets could

have reshaped the planetary architecture [53, 54]. Five- or six-planet versions invoke the ejection of an additional planet, often an ice giant [55, 56]. Other work places the instability much earlier, at the dispersal of the gaseous disk, and shows that growing terrestrial planets can be sculpted by such an instability [57, 58]. The theory therefore does not choose, at this stage, between an early and a later instability: it only requires that the Earth– B passage belong to B ’s history before its elimination.

Physical idea

Planet B is not an artificial object added only to the Earth–Moon problem. It may be understood as a temporary member of the primordial Solar System: it passes near the proto-Earth, reorganizes the circumterrestrial reservoir, and then rejoins the global heliocentric dynamics that removes part of the excess planetary population. The grazing passage and the later disappearance of B thus become two possible stages of the same history.

The complete path then takes the form

$$\begin{aligned} B_{\text{proto}} &\longrightarrow \text{grazing passage near the proto-Earth} \longrightarrow \mathcal{D}_1 \\ &\longrightarrow \text{heliocentric diffusion} \longrightarrow \text{elimination of } B. \end{aligned} \quad (1.5)$$

This extension does not replace the central mechanism of the theory. It gives it a wider logic: the secondary disk and orbital sorting explain lunar formation, while primordial orbital instability provides a natural fate for the transient agent B .

1.6 Scope of the theory

This theory is not presented as the unique possible history of lunar formation. Its objective is more limited and more precise: to determine whether the proposed physical path defines a parameter domain compatible with the full set of currently known dynamical, geochemical, isotopic, thermodynamic and chronological constraints.

The observations are therefore not treated as presumed validations, but as independent tests that can confirm, restrict or falsify the plausibility domain of the model.

Thus, even if all these constraints were satisfied simultaneously, the result would establish the existence of a viable and quantitatively coherent physical path among possible scenarios for the formation of the Moon, without claiming to demonstrate a unique history.

Physical idea

The object of the theory is the physical path. The disk is an initial condition, B is the agent of the transformation, the fluxes between reservoirs are the language, and geochemistry and dynamics are the tests. Each part of the work answers one aspect of the same question: *how could the observed Earth–Moon system have emerged through a physically coherent evolution?*

1.7 Status of the elements of the path

The proposed path combines elements whose epistemic status differs, and it is useful to distinguish them at the outset: one is borrowed from an already established framework, the other is the theory’s own and constitutes what it asks to be examined.

The *secondary disk* belongs to an established framework. The idea that a circumterrestrial reservoir contemporaneous with accretion could be the site of lunar formation is the idea of co-accretion models [22, 23], and the general principle of secondary accretion is illustrated by natural satellite systems (Chapter 2).

The *grazing non-collisional passage of a planet B* is the theory’s own element. The disk exists before the encounter; B therefore does not have to create it, and its action is limited to acting on

it through its gravitational field. This economy of means is the distinctive trait of the proposed path: the matter is already there, the angular momentum is already there, and the external agent merely redistributes both.

Physical idea

[The principle of precise circumstances] The Moon exists, and it has the particular properties observation ascribes to it: an unusual mass for a satellite of a terrestrial planet, an iron deficit, an isotopic composition indistinguishable from that of the terrestrial mantle, a high angular momentum. These properties do not follow from a general law; they call for *precise circumstances*. The proposed work therefore consists less in recounting what happened than in delimiting, in parameter space, the region where these circumstances are simultaneously realizable. That region is the *viable domain* $\mathcal{D}_{\text{viable}}$ defined in Chapter 15. If it is non-empty, the proposed path is physically practicable; if it is empty, it is not. It is in this sense that the theory is a path theory, and in this sense too that it is falsifiable.

Physical reading

In the proposed path, a gravitational interaction without contact suffices to reorganize an already constituted reservoir. B retains its physical identity and passes by. Its passage introduces an external torque able to redistribute the angular momentum of the preexisting reservoir.

1.8 Organization of the work

The work follows the physical path in the order imposed by the question (Figure 1.2). Part II establishes the astrophysical foundations: secondary accretion around planets is a real and recurrent phenomenon. Part III defines the four reservoirs and the laws governing their exchanges. Part IV orders the path by its time scales. Part V describes the action of B and its central result, orbital sorting. Part VI follows the circumterrestrial reservoir before and after the passage. Part VII organizes geochemical tests by families of tracers. Part VIII gathers the parameters, gates and numerical program. Part IX states the predictions in the form: *if the theory is correct... here is how to verify it*. The heliocentric fate of B runs through these parts as a coherence constraint: the gravitational agent must be able to pass near Earth and then be subsequently diffused or eliminated without remaining in the present planetary architecture.

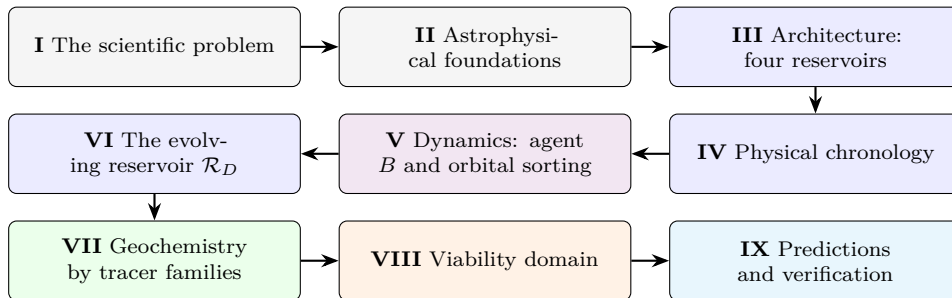


Figure 1.2: Architecture of the work: each part answers one aspect of the same question.

Part II

Astrophysical foundations

Before proposing a path for the Earth, the first element of that path must be shown to belong to known physics. This part shows that secondary accretion around planets is a real and recurrent phenomenon, while specifying what the analogies do and do not allow. It ends with the specificity of the terrestrial case: a solid-dominated disk whose feeding mode differs from that of giant planets.

Chapter 2

The principle of secondary accretion

2.1 The circumplanetary principle

Natural satellite systems provide a direct anchor for the notion of a secondary accretion system. The large regular satellites of Jupiter, Saturn and Uranus are mostly prograde, weakly eccentric and organized around their planet’s equatorial plane or Laplace plane. They are dynamical archives of circumplanetary evolution [15, 17, 20]. The following systems are retained as reference examples.

Table 2.1: Satellite systems used as support for the principle of circumplanetary accretion. The comparison concerns the physical principle of the disk, not an identity of mass or thermodynamics with the proto-Earth.

Planet	Satellites considered
Jupiter	Io, Europa, Ganymede, Callisto
Saturn	Mimas, Enceladus, Tethys, Dione, Rhea, Titan, Iapetus
Uranus	Miranda, Ariel, Umbriel, Titania, Oberon

Models for the formation of regular satellites explicitly treat matter supply, angular-momentum transport, cooling, migration and accretion within a disk surrounding the central planet [14, 10, 15].

Physical idea

The theory does not transpose to the proto-Earth the disk mass of Jupiter or Saturn. It retains a more fundamental result: a growing planetary body may be surrounded by a rotating reservoir that transports mass and angular momentum and can become the site of satellite accretion.

2.2 Physical meaning of the term “secondary”

The disk is called secondary relative to the primary circumsolar disk. The hierarchy of the two accretion systems is

$$\text{primary circumsolar disk} \longrightarrow \text{proto-Earth} \longrightarrow \text{secondary circumterrestrial disk.} \quad (2.1)$$

The word “secondary” describes a gravitational and accretional hierarchy. It does not mean that the disk appears late, nor that it is created by planet B .

2.3 Three nested propositions

The theory distinguishes three logical levels in order to separate general disk physics, what natural systems show, and what constitutes the specifically terrestrial proposition.

Principle of secondary circumplanetary accretion

P1 — General physics. A growing planet can receive matter carrying angular momentum and organize a fraction of it into a rotating circumplanetary reservoir. This reservoir transports mass and angular momentum and can become a site of satellite accretion [6, 14, 10].

P2 — Natural manifestations. The regular systems of Jupiter, Saturn and Uranus are natural expressions of secondary circumplanetary accretion. Their detailed histories differ from one system to another, but their architecture requires that matter bound to the planet was organized and accreted in its orbital environment [15, 17, 18].

P3 — Terrestrial proposition. The proto-Earth possessed a secondary circumterrestrial disk when planet B grazed it without collision. This encounter reorganized the reservoir and prepared the accretion of a dominant satellite: the Moon.

This separation protects the logic of the theory. The Jovian, Saturnian and Uranian systems show several physical realizations of the same general principle; the terrestrial branch must then be computed with the parameters proper to the proto-Earth and with the specific perturbation by B .

2.4 A hierarchy of accretion systems

The general hierarchy is

$$\text{circumstellar disk} \rightarrow \text{growing planet} \rightarrow \text{circumplanetary disk} \rightarrow \text{satellites} \quad (2.2)$$

The proposed terrestrial case is

$$\text{circumsolar disk} \rightarrow \text{proto-Earth} + \text{secondary disk} \xrightarrow{\text{passage of } B} \text{reorganized disk} \rightarrow \text{Moon}. \quad (2.3)$$

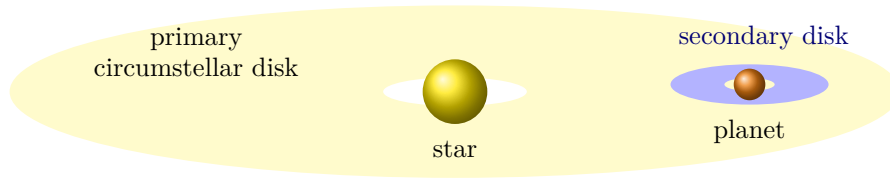


Figure 2.1: Principle of secondary accretion: the circumplanetary disk is an accretion system nested inside the primary circumstellar system.

2.5 Jupiter: the Galilean system as a dynamical reference

Io, Europa, Ganymede and Callisto form a compact system of prograde satellites with eccentricities of order 10^{-3} to 10^{-2} and small inclinations relative to Jupiter's Laplace plane [20]. This organization is consistent with accretion in a dissipative disk able to damp random motions.

The recent review of the Galilean system [15] identifies as major uncertainties the distribution of angular momentum in the matter feeding the disk, the origin of solids, and the thermal and viscous structure of the disk. These are precisely the quantities that describe the circumterrestrial disk:

$$\mathcal{V}_D = \left\{ \dot{M}_{\text{in}}, \Sigma(r, t), T(r, t), \nu(r, t), L_D(t), \mathbf{X}(r, t) \right\}. \quad (2.4)$$

The density gradient of the Galilean satellites, with an ice fraction increasing outward, shows that a satellite-forming disk imprints radial thermochemical information on the bodies that form within it. The function $T(r, t)$ must therefore be treated as a variable of material formation, not only as a dynamical parameter.

2.6 Saturn: viscous transport, Roche limit and satellite birth

Large moons such as Titan and Iapetus can be related to an ancient circumplanetary phase; Iapetus, which is more inclined than the inner satellites, is retained as a dynamically distinct limiting case. The ring environment illustrates another mechanism: a particulate reservoir spreads viscously and forms small satellites when a fraction of its matter crosses the Roche limit [16, 17]:

$$\text{circumplanetary reservoir} \rightarrow \text{spreading} \rightarrow \text{Roche crossing} \rightarrow \text{satellite accretion} \quad (2.5)$$

This sequence is directly useful for the terrestrial case: the Roche limit does not fix the inner edge of the disk; it marks a transition in the ability of matter to build durable self-gravitating aggregates.

2.7 Uranus: a regular system with a still-debated reservoir origin

Miranda, Ariel, Umbriel, Titania and Oberon form a regular system organized around Uranus's equator. The exact origin of this reservoir remains debated: models of a primordial circumplanetary disk and of a disk produced later have been proposed [18, 17]. Uranus is therefore used as an example of the accretion principle in a circumplanetary reservoir, without imposing an identity of origin with the terrestrial reservoir.

2.8 Neptune: Triton's capture, debris disk and reaccumulation

Neptune does not provide the same type of evidence as Jupiter or Saturn. Its dominant satellite, Triton, is retrograde and strongly inclined; it is therefore generally interpreted as a captured body, probably through a gravitational encounter involving a trans-Neptunian binary [48]. This capture profoundly reworked the Neptunian system.

The important point for the present theory is not Triton itself, but Neptune's inner satellites. In the scenario of Banfield and Murray, Triton's highly eccentric post-capture orbit excites the former inner satellites, drives collisions, forms a debris disk, and then leaves an inner system of satellites to reform on nearly equatorial orbits [49]. The discovery and analysis of Hippocamp reinforce the picture of a young, fragmented and reassembled inner system, strongly shaped by collisions [50]. Simulations of satellite formation around ice giants further show that Uranus and Neptune can form circumplanetary disks capable of producing regular satellite systems; Neptune may therefore have possessed an initially Uranus-like system before Triton's reworking [51]. Recent JWST observations of Neptune's rings and inner moons even suggest that their material may come from the interiors of older icy bodies that were destroyed or reworked [52].

Physical idea

The Neptunian case provides an analogy different from those of Jupiter and Saturn. It shows that an external gravitational event can destroy, sort or reorganize circumplanetary matter, then leave a reservoir from which inner satellites reform. In the proposed terrestrial path, *B* neither captures the Moon nor supplies its main mass; nevertheless it plays the analogous role of a gravitational agent of reorganization.

2.9 Direct observation: the circumplanetary disk of PDS 70c

ALMA has resolved around the young planet PDS 70c a compact dust source spatially separated from the circumstellar disk and interpreted as a circumplanetary disk [19]. The system provides a direct observation of two nested levels of accretion:

$$\text{circumstellar disk of PDS 70} \supset \text{PDS 70c} + \text{circumplanetary disk.} \quad (2.6)$$

Physical idea

PDS 70c shows that a planet still in formation can possess its own disk inside a larger circumstellar disk. The theory transposes this hierarchy principle to the proto-Earth, then adds the perturbation specific to it: the non-collisional passage of B .

Chapter 3

From the principle to the terrestrial case

Natural systems establish the principle; they do not fix terrestrial values. This chapter quantitatively compares satellite systems, formulates the constraint of a dominant satellite, and identifies the nature of the terrestrial disk.

3.1 Quantitative comparison of natural systems

The satellite mass fraction is defined as

$$\mu_{\text{sat}} = \frac{\sum_i M_i}{M_p}. \quad (3.1)$$

Using the gravitational parameters from JPL [20], one obtains the values in Table 3.1.

Table 3.1: Mass fraction of the selected satellite systems, computed from GM parameters (independent of the adopted value of G).

System	Satellites included	μ_{sat}
Jupiter	Io, Europa, Ganymede, Callisto	2.07×10^{-4}
Saturn	Mimas, Enceladus, Tethys, Dione, Rhea, Titan, Iapetus	2.47×10^{-4}
Uranus	Miranda, Ariel, Umbriel, Titania, Oberon	1.04×10^{-4}
Earth	Moon	1.23×10^{-2}

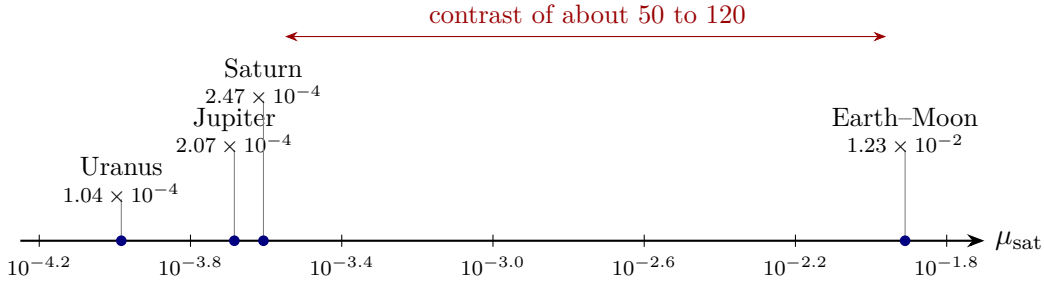


Figure 3.1: Mass contrast between the regular systems of the giant planets and the Earth–Moon system (logarithmic scale). This contrast is a quantitative constraint on the terrestrial case, not an objection to the circumplanetary principle.

The three giant-planet systems lie at order 10^{-4} , consistent with the regulation regime described for continuously fed gaseous disks [10]. The mass contrast is defined as

$$C_M = \frac{M_L/M_{\oplus}}{\mu_{\text{sat,giant}}}, \quad 50 \lesssim C_M \lesssim 120. \quad (3.2)$$

Quantitative control

This difference does not refute the principle of secondary accretion. It defines a constraint specific to the terrestrial case: the circumterrestrial disk, reorganized by B , must produce a satellite mass outcome far more concentrated than the regular systems of the giant planets. The theory must quantitatively explain why a lunar-mass dominant satellite emerges from the terrestrial system.

Table 3.2: Contrasts between the circumplanetary disks of the giants and the proposed circumterrestrial reservoir. The differences are not anomalies to be corrected: they define the problem the theory seeks to solve.

	Jupiter, Saturn	Earth (proposed)
Disk mass	$\sim 10^{-4} M_p$	$\sim 10^{-2} M_p$, to be explained
Dominant phase	gas and solids	solids only
Pressure support	yes, provided by gas	no
Lifetime	millions of years	a few years, Eq. (5.10)

Physical idea

[What orbital sorting comes to replace] In a circumjovian disk it is the gas that supplies pressure support, viscosity and duration: it leaves the system millions of years to transport angular momentum slowly outward and form satellites. The proposed circumterrestrial reservoir has none of this. It is solid-dominated, without pressure support, and its lifetime is counted in years. It therefore needs a mechanism that accomplishes at once what the gas accomplishes slowly. This is the function we propose to attribute to the passage of B : orbital sorting is the impulsive equivalent of viscous transport. The four contrasts of Table 3.2 are therefore not weaknesses of the analogy; they are the reason for the proposed mechanism.

3.2 From several satellites to one dominant satellite

A simple measure of satellite mass concentration is

$$f_{\text{dom}} = \frac{M_{\text{max}}}{\sum_i M_i}. \quad (3.3)$$

In the present terrestrial system, $f_{\text{dom}} \simeq 1$. The specific dynamical role of B is expressed as

$$\mathcal{D}_0 \xrightarrow{B} \mathcal{D}_1 \xrightarrow{\text{dissipation} + \text{accretion}} \{M_L, f_{\text{dom}} \simeq 1\}. \quad (3.4)$$

Falsification criterion

If, in the parameter domain that respects the other constraints, the post-encounter disk systematically produces several satellites of comparable mass or a total satellite mass far below the lunar mass, the proposed reorganization mechanism is insufficient.

3.3 Specificity of the terrestrial case: a solid-dominated disk

Jupiter, Saturn and PDS 70c exceed the *thermal mass* of their disk,

$$M_{\text{th}} \simeq \left(\frac{H}{r}\right)^3 M_{\odot} \approx 10 M_{\oplus} \quad \text{at 1 AU}, \quad (3.5)$$

beyond which captured gas forms a rotating disk. The proto-Earth, ten times less massive, lies in the *sub-thermal* regime. Its Bondi radius,

$$R_B = \frac{GM_{\oplus}}{c_s^2} \approx 4.0 \times 10^8 \text{ m} \approx 63 R_{\oplus} \quad (T \approx 280 \text{ K}, \mu = 2.34), \quad (3.6)$$

is smaller than its Hill radius ($\approx 235 R_{\oplus}$), and therefore fixes the accretion zone; high resolution simulations show that pebbles crossing the Bondi sphere of a rocky embryo are indeed accreted [21].

The circularization radius of matter with specific angular momentum $j \simeq \Omega R_{\text{acc}}^2/4$ is $r_c = j^2/(GM_{\oplus})$ [24]:

$$r_c(R_{\text{acc}} = R_H) = \frac{R_H}{48} \approx 4.9 R_{\oplus}, \quad r_c(R_{\text{acc}} = R_B) = \left(\frac{R_B}{R_H}\right)^4 \frac{R_H}{48} \approx 0.03 R_{\oplus}. \quad (3.7)$$

Nebular gas captured through the Bondi sphere therefore falls back onto the planet rather than forming a disk, in agreement with simulations of sub-thermal embryo envelopes supported by pressure rather than rotation [26, 27]. The terrestrial secondary disk is therefore dominated by solids, with three feeding channels:

$$\dot{M}_{\text{in}} = \dot{M}_{\text{coacc}} + \dot{M}_{\text{ej}} + \dot{M}_{\text{rot}}, \quad (3.8)$$

where $\dot{M}_{\text{coacc}}$ represents capture by collisions of planetesimals within the Hill sphere [22, 23], $\dot{M}_{\text{ej}}$ ordinary accretion-impact ejecta, and $\dot{M}_{\text{rot}}$ equatorial mass loss from a proto-Earth rotating close to the critical limit.

Physical reading

The analogy with giant planets concerns the principle of a circumplanetary reservoir, not its mode of supply. The rotational channel $\dot{M}_{\text{rot}}$ is the one most directly coupled to the rest of the theory: it supplies material of terrestrial silicate composition, oriented by the spin, and links the fast rotation of the proto-Earth to the very existence of the disk.

Didactic reminder

Part II has established two results. First, a rotating circumplanetary reservoir is a real astrophysical object. Second, in the terrestrial case this reservoir is dominated by solids and must produce a satellite mass concentrated in one body. The next part translates these results into a common physical language: reservoirs and fluxes.

Part III

Physical architecture: the four reservoirs

The physical path is described as a sequence of exchanges between reservoirs. This language has a decisive advantage: it requires the balances of mass, angular momentum, energy and composition to be closed at every step, and it gives the role of B a precise formulation. This part defines the reservoirs, the fluxes connecting them and the conservation laws that constrain them, then fixes the Level I state variables and numerical reference scales.

Chapter 4

Reservoirs, fluxes and conservation laws

4.1 The four reservoirs of the system

The proto-terrestrial system is decomposed into four reservoirs:

$$\mathcal{R} = \{\mathcal{R}_\oplus, \mathcal{R}_D, \mathcal{R}_B, \mathcal{R}_\infty\}. \quad (4.1)$$

Definition

$\mathcal{R}_\oplus$ — the proto-Earth: its mass, spin, internal structure and the composition of its layers.
 $\mathcal{R}_D$ — the secondary circumterrestrial reservoir: gaseous, multiphase or particulate disk, and the population of aggregates or moonlets it contains.
 $\mathcal{R}_B$ — planet B : its mass, spin and orbital motion relative to the proto-Earth.
 $\mathcal{R}_\infty$ — the external medium not bound to the Earth–disk system: heliocentric matter entering the system (co-accretion) or leaving it (escape).

Each reservoir i is described by a state

$$\mathcal{X}_i = \{M_i, \mathbf{J}_i, E_i, \mathbf{X}_i\}, \quad (4.2)$$

which gathers its mass, angular momentum, energy and composition. Angular momenta are defined in the barycentric frame of the system:

$$\mathbf{J}_\oplus = \mathbf{S}_\oplus, \quad \mathbf{J}_D = \mathbf{L}_D + \sum_k \mathbf{L}_k, \quad \mathbf{J}_B = \mathbf{S}_B + \mathbf{L}_{\text{orb}}, \quad \mathbf{J}_\infty = \mathbf{L}_{\text{esc}}, \quad (4.3)$$

where $\mathbf{L}_k$ is the orbital angular momentum of moonlets and $\mathbf{L}_{\text{orb}}$ that of the relative Earth– B motion.

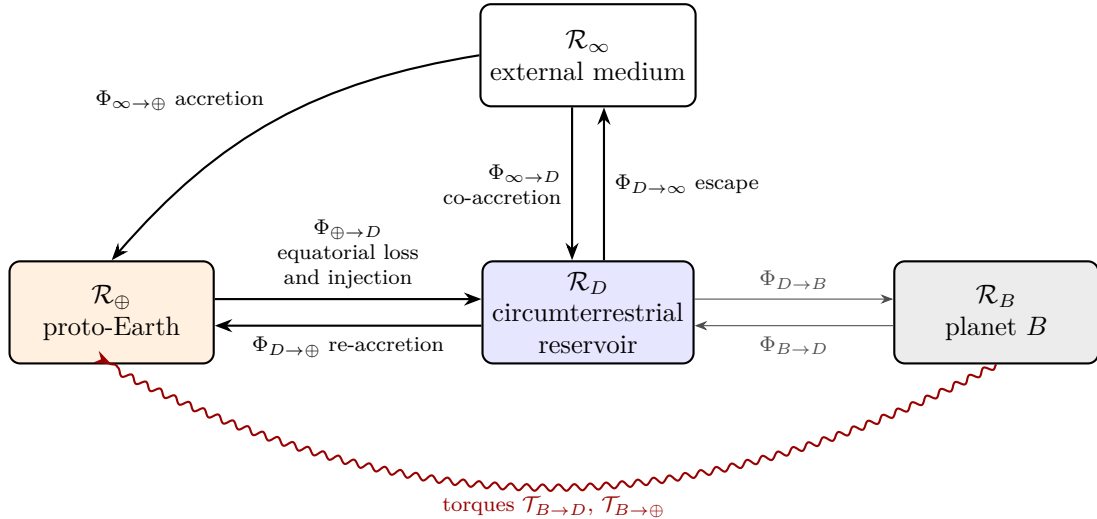


Figure 4.1: The four reservoirs and their exchanges. Solid arrows represent mass fluxes, carrying angular momentum, energy and composition; the wavy arrow represents the gravitational torques exerted by B during its passage.

4.2 Mass fluxes

Let $\Phi_{i \rightarrow j} \geq 0$ denote the mass flux from reservoir i to reservoir j . The evolution of each reservoir is

$$\frac{dM_i}{dt} = \sum_{j \neq i} (\Phi_{j \rightarrow i} - \Phi_{i \rightarrow j}), \quad \sum_i M_i = \text{constant}. \quad (4.4)$$

The principal fluxes are grouped in Table 4.1. Equation (10.2) for the disk is the special case $i = D$ of this relation.

Table 4.1: Principal mass fluxes between reservoirs and associated processes.

Flux	Process	Dominant phase
$\Phi_{\infty \rightarrow \oplus}$	growth of the proto-Earth	growth
$\Phi_{\infty \rightarrow D}$	co-accretion, capture in the Hill sphere	growth
$\Phi_{\oplus \rightarrow D}^{\text{rot}}$	rotational equatorial mass loss	growth
$\Phi_{\oplus \rightarrow D}^{\text{mar}}$	injection by the tide of B	encounter
$\Phi_{D \rightarrow \oplus}$	re-accretion	encounter, relaxation
$\Phi_{D \rightarrow \infty}$	escape	encounter, relaxation
$\Phi_{D \rightarrow B}, \Phi_{B \rightarrow D}$	capture by B , material ceded by B	encounter

4.3 Angular momentum

A mass flux carries the specific angular momentum $\mathbf{j}_{i \rightarrow j}$ of the transferred matter, $\mathbf{\Lambda}_{i \rightarrow j} = \Phi_{i \rightarrow j} \mathbf{j}_{i \rightarrow j}$. Reservoirs also exchange angular momentum through gravitational torques $\mathcal{T}_{j \rightarrow i}$:

$$\frac{d\mathbf{J}_i}{dt} = \sum_{j \neq i} (\mathbf{\Lambda}_{j \rightarrow i} - \mathbf{\Lambda}_{i \rightarrow j}) + \sum_{j \neq i} \mathcal{T}_{j \rightarrow i}, \quad \mathcal{T}_{j \rightarrow i} = -\mathcal{T}_{i \rightarrow j}. \quad (4.5)$$

Internal torques cancel pairwise, so that

$$\boxed{\sum_i \mathbf{J}_i = \text{constant}} \quad (4.6)$$

as long as the solar torque is negligible, which is the case during the encounter. The angular momentum balance of the encounter (Chapter 9) is the direct application of this law.

Physical idea

In this language, the role of B becomes exact. During its passage, the torques $\mathcal{T}_{B \rightarrow D}$ and $\mathcal{T}_{B \rightarrow \oplus}$ dominate the other terms, and they trigger the fluxes $\Phi_{\oplus \rightarrow D}$, $\Phi_{D \rightarrow \oplus}$ and $\Phi_{D \rightarrow \infty}$. When B moves away, these terms vanish: this is the meaning of *transient* in the definition of B .

4.4 Energy

Mechanical energy is not conserved along the path: the relaxation of the reservoir converts orbital energy into heat, which is then radiated. The global balance is

$$\frac{d}{dt} \left(\sum_i E_i + E_{\text{int}} + E_{\text{th}} \right) = -\mathcal{L}_{\text{rad}}, \quad (4.7)$$

where E_{int} is the gravitational interaction energy between reservoirs, E_{th} the thermal energy, and $\mathcal{L}_{\text{rad}}$ the radiated power. Dissipation by shocks, collisions and viscosity, detailed in Chapter 12, is the source term for E_{th} .

4.5 Composition

Each flux carries a composition $\mathbf{X}_{i \rightarrow j}$ that is not necessarily that of the mean reservoir: it depends on the depth, latitude and phase of the transferred matter. The composition of each reservoir evolves according to

$$\frac{d(M_i \mathbf{X}_i)}{dt} = \sum_{j \neq i} (\Phi_{j \rightarrow i} \mathbf{X}_{j \rightarrow i} - \Phi_{i \rightarrow j} \mathbf{X}_{i \rightarrow j}) + \mathbf{\Pi}_i, \quad (4.8)$$

where $\mathbf{\Pi}_i$ represents internal transformations within the reservoir: melting, condensation, evaporation, metal segregation.

Physical reading

Equation (4.8) is the entry point of geochemistry. The flux $\Phi_{\oplus \rightarrow D}^{\text{mar}}$ samples matter from a specific layer of the proto-Earth; its composition $\mathbf{X}_{\oplus \rightarrow D}$ therefore carries the memory of the latitude and depth from which it was stripped. Part VII uses this memory.

4.6 Level I state variables

Level I characterizes the admissible state of the Earth–disk system just before the passage of B , then measures its transformation during the encounter. Natural satellite systems define classes of behavior to reproduce or distinguish: angular-momentum transport, cooling, crossing of the Roche limit, migration, accretion and concentration of satellite mass.

The proto-terrestrial state is

$$\Psi_{\oplus} = \{M_{\oplus}, R_{\text{eq}}, \Omega_{\oplus}, S_{\oplus}, T_{\oplus}(r), \rho_{\oplus}(r), \mathbf{X}_{\oplus}(r)\}, \quad (4.9)$$

and the disk state is

$$\mathcal{D}_0 = \left\{ M_{D,0}, \Sigma_0(r), R_{\text{in}}, R_{\text{out}}, T_0(r), P_0(r), H_0(r), \right. \\ \left. L_{D,0}, v_{r,0}(r), \kappa_0(r), \mathbf{X}_0(r), \mathbf{f}_{\text{phase},0}(r), \mathcal{G}_0 \right\}, \quad (4.10)$$

where $\mathcal{G}_0 = \{m_k, a_k, e_k, i_k, \mathbf{X}_k\}_{k=1}^{N_0}$ describes the population of aggregates or moonlets already formed at t_B (Section 5.5). The vector of phase fractions is

$$\mathbf{f}_{\text{phase}} = (f_{\text{vap}}, f_{\text{liq}}, f_{\text{sol}}), \quad f_{\text{vap}} + f_{\text{liq}} + f_{\text{sol}} = 1. \quad (4.11)$$

4.7 Encounter parameters

The state of B is described by

$$\Psi_B = \{M_B, R_B, \rho_B, q_B, v_{\infty}, i_B, \omega_B, \varpi_B, \mathbf{S}_B\}, \quad (4.12)$$

where q_B is the minimum geocentric distance of the nominal trajectory, v_{∞} the relative asymptotic velocity, i_B the inclination of the trajectory relative to the disk plane, ω_B the argument of periastron measured from the line of nodes, ϖ_B the longitude of periastron, and $\mathbf{S}_B$ the spin of B . The minimal parameter vector is

$$\Theta_I = \{\Psi_{\oplus}, \mathcal{D}_0, \Psi_B, t_B\}. \quad (4.13)$$

To connect the encounter to the later fate of B , we also define the heliocentric extension

$$\Theta_I^+ = \{\Theta_I, \mathcal{H}_{B,1}, t_{\text{ej}}\}, \quad \mathcal{H}_{B,1} = \{a_{B,1}, e_{B,1}, i_{B,1}, \Omega_{B,1}, \omega_{B,1}\}, \quad (4.14)$$

where $\mathcal{H}_{B,1}$ denotes the heliocentric elements of B after its passage near the proto-Earth. This extension is not required to compute the local orbital sorting, but it becomes necessary to test the connection with the global instability of the Solar System.

Falsification criterion

Level I fails if no continuous domain of Θ_I simultaneously allows a physically persistent preexisting disk, a non-collisional encounter, a closed angular-momentum balance, sufficient post-encounter disk mass and a composition compatible with lunar constraints.

4.8 Reference scales of the proto-Earth–disk– B system

Table 4.2 gathers the numerical scales that bound Level I. They are computed for a proto-Earth with present terrestrial mass and radius at 1 AU; the values associated with B correspond to a reference case $M_B = 0.1 M_\oplus$, $q_B = 2 R_\oplus$, parabolic trajectory.

Table 4.2: Reference scales. These values are not solutions: they define the orders of magnitude against which the outputs of the calculation must be placed.

Quantity	Expression	Value
Hill radius	$a_\oplus (M_\oplus / 3M_\odot)^{1/3}$	$1.50 \times 10^9 \text{ m} \approx 235 R_\oplus$
Truncated outer edge	$\xi_H R_H$, $\xi_H \approx 0.3\text{--}0.4$	$\approx 70\text{--}95 R_\oplus$
Thermal mass	$(H/r)^3 M_\odot$	$\approx 10 M_\oplus$
Bondi radius	$GM_\oplus / c_s^2$	$\approx 63 R_\oplus$
Circularization (Hill)	$R_H / 48$	$\approx 4.9 R_\oplus$
Circularization (Bondi)	$(R_B / R_H)^4 R_H / 48$	$\approx 0.03 R_\oplus$
Roche limit ($\rho_s = 3.34 \text{ g cm}^{-3}$)	Eq. (10.10)	$1.85 \times 10^7 \text{ m} \approx 2.9 R_\oplus$
Orbital period at a_{Roche}	$2\pi (a_{\text{Roche}}^3 / GM_\oplus)^{1/2}$	$\approx 7.0 \text{ h}$
Synchronous orbit ($P_\oplus = 4\text{--}5 \text{ h}$)	$(GM_\oplus / \Omega_\oplus^2)^{1/3}$	$\approx 2.0\text{--}2.3 R_\oplus$
Critical rotation period	$2\pi (R_\oplus^3 / GM_\oplus)^{1/2}$	$\approx 1.4 \text{ h}$
Critical spin ($k \approx 0.33$)	$k M_\oplus R_\oplus^2 \Omega_c$	$\approx 1.0 \times 10^{35} \text{ kg m}^2 \text{ s}^{-1}$
Present Earth–Moon angular momentum	L_{EM}	$\approx 3.5 \times 10^{34} \text{ kg m}^2 \text{ s}^{-1}$
Orbital angular momentum of one Moon at a_{Roche}	$M_L (GM_\oplus a_{\text{Roche}})^{1/2}$	$\approx 6.3 \times 10^{33} \text{ kg m}^2 \text{ s}^{-1}$
Relative orbital angular momentum of B	Eq. (6.18)	$\approx 5.7 \times 10^{34} \text{ kg m}^2 \text{ s}^{-1}$
Periastron velocity of B	$(2GM_{\text{tot}} / q_B)^{1/2}$	$\approx 8.3 \text{ km s}^{-1}$
Characteristic encounter duration	q_B / v_p	$\approx 26 \text{ min}$
Roche distance of B ($\rho_B = 3.93 \text{ g cm}^{-3}$)	Eq. (6.21)	$\approx 2.75 R_\oplus$

Quantitative control

Three comparisons structure what follows. The relative orbital angular momentum carried by B is of the same order as the total angular momentum of the Earth–Moon system: without collision, a grazing passage has an angular-momentum reservoir on the scale of the problem. The encounter duration is short compared with the orbital period at the Roche limit: the disk undergoes an impulsive forcing. Finally, the Roche distance of B is close to the passage distances considered: the integrity of B becomes an explicit constraint.

Part IV

Physical chronology

The chronology is not narrated here as a sequence of episodes. It is constructed from physics: each phase of the path is defined by the terms that dominate the flux equations (4.4)–(4.8), and by its own time scale. The separation of these scales justifies dividing the calculation into successive phases.

Chapter 5

Physical chronology of the path

5.1 Phases defined by dominant fluxes

Table 5.1 defines the phases of the path by their dominant terms.

Table 5.1: Phases of the physical path, defined by the dominant terms of the flux equations.

Phase	Dominant terms	Time scale
0 — Growth and feeding	$\Phi_{\infty \rightarrow \oplus}$, $\Phi_{\infty \rightarrow D}$, $\Phi_{\oplus \rightarrow D}^{\text{rot}}$, torques $\mathcal{T}_{\oplus \rightarrow D}$	10^6 – 10^8 yr
I — Pre-encounter state	viscous transport, accretion into aggregates and moonlets	t_ν , t_{sat}
II — Encounter	$\mathcal{T}_{B \rightarrow D}$, $\mathcal{T}_{B \rightarrow \oplus}$, $\Phi_{\oplus \rightarrow D}^{\text{mar}}$	$t_{\text{enc}} \sim 0.5$ – 2 h
III — Sorting and relaxation	$\Phi_{D \rightarrow \oplus}$, $\Phi_{D \rightarrow \infty}$, dissipation	10 – 10^3 orbital periods
IV — Lunar accretion	accretion beyond Roche, transport from the inner disk	months to 10^2 yr
V — Lunar magma ocean	cooling and differentiation of the Moon	10^7 – 10^8 yr
VI — Fate of B	heliocentric diffusion, planetary encounters, elimination or ejection	early or late instability

5.2 Hierarchy of time scales

The characteristic time scales are ordered as

$$t_{\text{enc}} < P_{\text{orb}}(a_{\text{Roche}}) \ll t_{\text{relax}} \lesssim t_{\text{acc},L} \ll t_{\text{LMO}} \quad (5.1)$$

with $t_{\text{enc}} \approx 26$ min in the reference case and $P_{\text{orb}}(a_{\text{Roche}}) \approx 7$ h (Table 4.2). Figure 5.1 places these scales on a logarithmic axis.

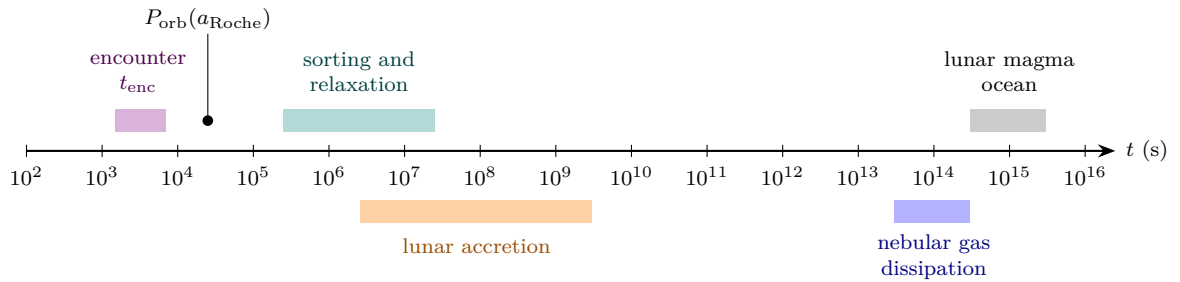


Figure 5.1: Time scales of the physical path (order-of-magnitude values). Eleven orders of magnitude separate the encounter duration from that of the lunar magma ocean.

Physical idea

The encounter lasts less than one orbital period at the Roche limit: during the passage, the reservoir does not have time to evolve viscously and may be treated as frozen. After the passage, relaxation extends over hundreds of periods, without further intervention by B . This separation of scales justifies the division of the numerical program into Phases A, B and C

(Chapter 16).

5.3 Order of stages

The minimal chronology is

$$t_{\text{accr},\oplus}^{\text{start}} < t_D^{\text{form}} < t_B < t_D^{\text{reorg}} < t_L^{\text{accr}} < t_{\text{LMO}}, \quad (5.2)$$

This chain fixes the internal order of the Earth–Moon problem. The heliocentric connection adds an independent condition:

$$\boxed{t_B < t_{\text{ej}}}, \quad (5.3)$$

without requiring t_{ej} to belong to either an early or a late instability. The calculation must simply show that a post-passage orbit of B can join a trajectory eliminated by global dynamics.

and a more complete chronological vector is

$$\mathbf{t} = \left\{ t_D^{\text{form}}, t_D^{\text{feed}}, t_D^{\text{cool}}, t_D^{\text{cond}}, t_B, t_{\text{relax}}, t_{\text{Roche}}, t_{\text{acc},L}^{\text{start}}, t_{\text{acc},L}^{\text{end}}, t_{\text{LMO}}, t_{\text{ej}} \right\}. \quad (5.4)$$

5.4 Gaseous disk and residual reservoir

The theory distinguishes the physical states of the circumterrestrial reservoir:

$$\mathcal{D}(t) = \mathcal{D}_{\text{gas}} \cup \mathcal{D}_{\text{multiphase}} \cup \mathcal{D}_{\text{particulate}}, \quad (5.5)$$

the weighting between these regimes being computed by $\mathbf{f}_{\text{phase}}(r, t)$.

Nebular gas dissipates within a few million years, whereas lunar chronometers place the accretion of the Moon several tens of millions of years after the formation of the first solids; rapid growth of the proto-Earth by pebble accretion [9] and its later completion bound the interval $[t_{\text{accr},\oplus}^{\text{start}}, t_B]$. At the time of the passage, the reservoir is therefore described by its multiphase and particulate components, fed by the channels of Eq. (3.8).

Physical reading

This distinction prevents the theory from depending on the persistence of a gas-rich disk. The reservoir evolves toward a multiphase or particulate state while conserving circumterrestrial mass and angular momentum, and it is this state that the passage of B reorganizes.

5.5 Three chronological regimes

Let t_{sat} be the characteristic time of satellite accretion and $\Delta t_{\text{preB}} = t_B - t_{D,\text{form}}$ the time available before the passage. We distinguish

$$\text{late accretion regime:} \quad t_{\text{sat}} \gg \Delta t_{\text{preB}}, \quad (5.6)$$

$$\text{intermediate regime:} \quad t_{\text{sat}} \sim \Delta t_{\text{preB}}, \quad (5.7)$$

$$\text{early accretion regime:} \quad t_{\text{sat}} \ll \Delta t_{\text{preB}}. \quad (5.8)$$

The available time scales orient this choice. An unfed gaseous component spreads in

$$t_\nu = \left[\alpha h^2 \Omega_K \right]^{-1} \approx 80 \text{ yr} \quad (\alpha = 10^{-3}, h = H/r = 0.1, r = 10 R_\oplus), \quad (5.9)$$

This value is evaluated at $10 R_\oplus$. It is useful to restate it at the radius where the reservoir actually lies, since $\Omega_K \propto r^{-3/2}$:

$$t_\nu \simeq 7 \text{ yr at } 2 R_\oplus, \quad t_\nu \simeq 13 \text{ yr at } 2.9 R_\oplus \quad (\alpha = 10^{-3}, h = 0.1), \quad (5.10)$$

and to note that for a massive reservoir, whose transport is dominated by gravitational torques, the effective α rises to 10^{-2} – 10^{-1} and reduces these durations to a few months. The time window offered to the passage is therefore a constraint in its own right, and Eq. (5.10) is the reference for gate (11.11). A particulate component located beyond the Roche limit aggregates in a few months to about a year [12, 31]. These reference times are compared with Δt_{preB} and the disk feeding rate; the calculation determines which of the three regimes belongs to the viable domain. When the reservoir contains moonlets $\mathcal{G}_0$ at the time of the passage, those located beyond a_{sync} migrate outward.

Physical reading

When moonlets are present, the role of B becomes more precise: by exciting eccentricities and inclinations, it produces orbit crossings between moonlets and between moonlets and the residual disk, and therefore mergers. The grazing passage becomes a mechanism for concentrating satellite mass, giving physical content to the condition $f_{\text{dom}} \rightarrow 1$.

Didactic reminder

The chronology has fixed the order and duration of the phases. The next part enters the shortest and most decisive phase: the encounter, during which agent B redistributes the mass and angular-momentum fluxes.

Part V

Dynamics: the gravitational agent B

This part is the original core of the theory. It describes how a planet B , without ever touching the proto-Earth, redistributes mass and angular-momentum fluxes between reservoirs. It contains four chapters: the passage itself and its low-latitude geometry, the injection of superficial proto-terrestrial matter, the central result of the passage — the orbital sorting that defines the lunar window — and finally the balances that close the encounter.

Chapter 6

The non-collisional passage

Consistently with its definition (Chapter 1), B acts on the Earth–reservoir system solely through its gravitational field. This chapter establishes the frequency of such passages, their geometry, their dynamical regime and the exact force field they exert.

6.1 A gravitational interaction without contact

During the growth of terrestrial planets, planetary embryos cross one another and undergo many close encounters. Including gravitational focusing, the cross-section for encounters whose periastron is smaller than q is

$$\sigma(q) = \pi q^2 \left[1 + \frac{2GM_{\text{tot}}}{q v_{\infty}^2} \right], \quad M_{\text{tot}} = M_{\oplus} + M_B. \quad (6.1)$$

Let $R_c = R_{\text{eq}} + R_{B,\text{eff}}$ be the contact distance and $\Theta = 2GM_{\text{tot}}/(R_c v_{\infty}^2)$ the focusing parameter. The ratio between the number of grazing passages ($R_c < q < 2R_c$) and collisions ($q < R_c$) is

$$\frac{\mathcal{N}_{\text{graze}}}{\mathcal{N}_{\text{coll}}} = \frac{\sigma(2R_c) - \sigma(R_c)}{\sigma(R_c)} = \frac{3 + \Theta}{1 + \Theta} \in [1, 3]. \quad (6.2)$$

Original contribution of the theory

Grazing passages are at least as frequent as collisions, and up to three times more frequent when gravitational focusing is weak. The mechanism of the theory therefore does not rely on an exceptional event: it selects, from the natural population of planetary encounters, those that do not end in contact. Non-collisional encounters are also recognized as an efficient process in early Earth–Moon dynamics [32]; the theory assigns them an earlier and more fundamental role here: reorganizing the disk from which the Moon forms.

6.2 Geometrical definition of the non-collisional character

The scenario is gravitational. The trajectory of B does not touch the proto-Earth:

$$q_B > R_{\text{eq}} + R_{B,\text{eff}}, \quad (6.3)$$

where R_{eq} is the equatorial radius of the rotating proto-Earth, larger than the polar radius, and $R_{B,\text{eff}}$ is the radius of B deformed by the terrestrial tide. The eccentricity of the relative hyperbolic trajectory is

$$e_B = 1 + \frac{q_B v_{\infty}^2}{GM_{\text{tot}}}. \quad (6.4)$$

The trajectory crosses the disk plane at two nodes located at

$$r_{\text{node}}^{\pm} = \frac{q_B (1 + e_B)}{1 \pm e_B \cos \omega_B}. \quad (6.5)$$

For configurations in which the theory requires B not to cross the dense part of the disk, one imposes

$$R_{\text{out,dense}} < \min(r_{\text{node}}^+, r_{\text{node}}^-) = \frac{q_B (1 + e_B)}{1 + e_B |\cos \omega_B|}. \quad (6.6)$$

For a quasi-parabolic trajectory whose periastron is highest above the disk plane ($\omega_B = \pi/2$), this bound is $2q_B$.

Physical reading

The action of B is carried by its gravitational field. The inclined geometry allows a strong perturbation of the disk while maintaining physical separation between B , the proto-Earth and the dense part of the disk: a dense disk extending over a few Earth radii, consistent with the circularization radius of Eq. (3.7), is entirely compatible with a passage at q_B of a few $R_\oplus$.

6.3 Dynamical regime of the encounter

The relative velocity at periastron and the characteristic duration of the encounter are

$$v_p = \sqrt{v_\infty^2 + \frac{2GM_{\text{tot}}}{q_B}}, \quad t_{\text{enc}} \simeq \frac{q_B}{v_p}. \quad (6.7)$$

The response regime of a disk region at radius r is measured by

$$\eta(r) = \Omega_K(r) t_{\text{enc}}. \quad (6.8)$$

For $\eta \ll 1$ the response is impulsive, for $\eta \sim 1$ it is resonant, and for $\eta \gg 1$ it is adiabatic and the forcing is attenuated. In the reference case of Table 4.2, $t_{\text{enc}} \approx 26$ min and $\eta(a_{\text{Roche}}) \approx 0.4$: the region where the Moon assembles receives an impulsive forcing. For the proto-Earth itself, the analogous parameter $\eta_\oplus = \Omega_\oplus t_{\text{enc}}$ is of order unity when the rotation is close to the critical limit: the tide of B then acts on a time scale close to the rotation period, which strengthens the coupling between the passage and equatorial matter.

6.4 Exact differential field

In a frame centered on the proto-Earth, the exact differential acceleration due to B on a parcel located at $\mathbf{r}$ is

$$\mathbf{a}_{B,\text{diff}} = -GM_B \left[\frac{\mathbf{r} - \mathbf{r}_B(t)}{|\mathbf{r} - \mathbf{r}_B(t)|^3} + \frac{\mathbf{r}_B(t)}{|\mathbf{r}_B(t)|^3} \right]. \quad (6.9)$$

The first term is the direct attraction by B , the second the acceleration of the proto-Earth itself toward B (indirect term). This expression is used without multipolar expansion in the grazing regime, where $|\mathbf{r}|$ and $|\mathbf{r}_B|$ are comparable.

6.5 Torque exerted on the disk

The total torque of B on the disk and the resulting change in angular momentum are

$$\mathcal{T}_{B \rightarrow D} = \int_D \mathbf{r} \times \mathbf{a}_{B,\text{diff}} dm, \quad \Delta \mathbf{L}_D = \int_{t_i}^{t_f} \mathcal{T}_{B \rightarrow D} dt + \Delta \mathbf{L}_{\text{flux}}, \quad (6.10)$$

where $\Delta \mathbf{L}_{\text{flux}}$ represents the angular momentum transported by matter that enters or leaves the disk during the encounter. The main dynamical outputs are

$$\mathcal{O}_B = \{\Delta M_D, \Delta \mathbf{L}_D, \Delta e_D, \Delta i_D, \Delta R_{\text{out}}, \Delta T_D, \Delta \mathcal{G}\}. \quad (6.11)$$

6.6 Low-latitude geometry of the passage

The geometry retained by the theory is a passage *parallel to the dynamical low-latitude band* of an oblate, rapidly rotating proto-Earth, in the direction of its rotation. It combines four favorable effects.

Targeted latitude. At periastron, the point of the proto-Earth vertically below B lies at latitude

$$\lambda_B = \arcsin(\sin i_B \sin \omega_B), \quad (6.12)$$

where i_B is the inclination of the trajectory relative to the rotational equator and ω_B the argument of periastron. The low-latitude passage condition is

$$|\lambda_B| \leq \lambda_c, \quad (6.13)$$

that is, $i_B \lesssim \lambda_c$ for $\omega_B = 90^\circ$. The quantity λ_c defines a dynamical low-latitude band around the instantaneous rotational equator of the proto-Earth. Its value is not imposed by the present obliquity; it is determined by the coupling between rotation, oblateness, the tide of B and the response of the reservoir.

Equatorial forcing is preserved. The radial tidal acceleration varies as $(3 \cos^2 \psi - 1)$, where ψ is the angle from the sub- B point. At the equator, at the longitude of the targeted point, $\psi = \lambda_B$ and the relative forcing is

$$F(\lambda_B) = \frac{3 \cos^2 \lambda_B - 1}{2}, \quad F(20^\circ) \approx 0.82. \quad (6.14)$$

A slight inclination therefore costs little equatorial forcing.

Oblateness strengthens the action of B . For an oblate proto-Earth, the equatorial effective gravity is, to first order in J_2 ,

$$g_{\text{eff,eq}} = \frac{GM_\oplus}{R_{\text{eq}}^2} \left(1 + \frac{3}{2} J_2 \right) - \Omega_\oplus^2 R_{\text{eq}}. \quad (6.15)$$

The injection threshold (Chapter 7) contains the factor $(R_{\text{eq}}/q_B)^3$: an equatorial radius larger by 10 to 20% increases the tidal term by 33 to 73%, precisely where the effective gravity is weakest.

The prograde sense prolongs the forcing. If B passes in the direction of the proto-Earth's rotation, the same longitude remains longer under the tidal bulge. With $\Omega_B = v_p/q_B$ the angular velocity of B at periastron, the ratio of the prograde and retrograde forcing durations is

$$\frac{\tau_{\text{pro}}}{\tau_{\text{retro}}} = \frac{\Omega_\oplus + \Omega_B}{\Omega_\oplus - \Omega_B}. \quad (6.16)$$

For $q_B = 3 R_\oplus$, $\Omega_B \approx 3.5 \times 10^{-4} \text{ rad s}^{-1}$; for a proto-Earth close to critical rotation, $\Omega_\oplus \approx 1.2 \times 10^{-3} \text{ rad s}^{-1}$, and the ratio is about 1.8. This mechanism is the planetary analogue of the efficiency of prograde encounters shown in galaxies [45].

The geometry remains compatible with non-crossing of the dense disk. With $\omega_B = 90^\circ$, B passes at periastron at height $q_B \sin i_B$ above the equatorial plane and crosses that plane only at nodes located at about $2q_B$ (Eq. (6.5)). A dense reservoir extending to $5 R_\oplus$ is not crossed once $q_B \gtrsim 2.5 R_\oplus$.

Effect on the reservoir. Scaling laws established for encounters between a particle disk and a perturber on a coplanar prograde parabolic trajectory indicate a truncation radius

$$r_{\text{tr}} \approx 0.28 q_B \left(\frac{M_B}{M_\oplus} \right)^{-0.32} \quad (6.17)$$

[33], or $r_{\text{tr}} \approx 0.6 q_B$ for $M_B = 0.1 M_\oplus$. The prograde low-latitude band is therefore the most active region of parameter space: it maximizes both injection and orbital sorting. The escaped fraction η_{esc} (Chapter 8) is the control variable that delimits favorable passages within this band.

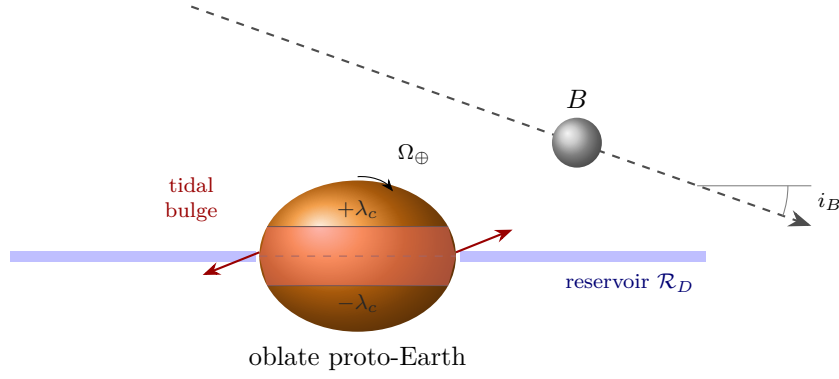


Figure 6.1: Low-latitude geometry: B passes on a trajectory weakly inclined with respect to the rotational equator of an oblate proto-Earth, in the direction of its rotation. The tidal bulge forms in the low-latitude band, where effective gravity is minimal; the dense reservoir remains inside the nodes of the trajectory (not to scale).

Physical idea

The stripped layer is the outermost and therefore the least dense: silicate already depleted in Fe–Ni metal and siderophile elements by core formation. This does not imply depletion in silicate FeO.

Observable consequence

The injected matter carries the signature of the low-latitude external silicate layer: depletion in metal, Ni, Co, W and V inherited from terrestrial core formation, and relative enrichment in incompatible elements if the layer is differentiated. The tracers in Part VII directly test this signature.

6.7 Angular-momentum reservoir carried by B

In the barycentric frame of the Earth– B pair, the relative orbital angular momentum of the encounter is

$$L_{B,\text{rel}} = \mu_r \sqrt{GM_{\text{tot}} q_B (1 + e_B)}, \quad \mu_r = \frac{M_{\oplus} M_B}{M_{\text{tot}}}. \quad (6.18)$$

For $M_B = 0.1 M_{\oplus}$, $q_B = 2 R_{\oplus}$ and $e_B = 1$, one obtains $L_{B,\text{rel}} \approx 5.7 \times 10^{34} \text{ kg m}^2 \text{ s}^{-1}$, about $1.6 L_{\text{EM}}$. The fraction exchanged with the Earth–disk system,

$$\chi_J = \frac{|\Delta J_{\oplus D}|}{L_{B,\text{rel}}}, \quad (6.19)$$

is an output of Phase B of the numerical program.

Original contribution of the theory

Without any collision, the trajectory of B carries an angular momentum of the same order as that of the Earth–Moon system. Through gravitational torque alone, the grazing passage can therefore play the role assigned by impact scenarios to the collision: setting the angular-momentum budget of the Earth–disk system. The exchange can occur in either direction; if the rapidly rotating proto-Earth carries a spin larger than L_{EM} (Table 4.2), the trajectory of B can receive part of it.

6.8 Integrity of B during the encounter

The terrestrial tide at the surface of B is characterized by

$$\tau_B = 2 \frac{M_\oplus}{M_B} \left(\frac{R_B}{d} \right)^3, \quad (6.20)$$

the ratio between tidal acceleration and self-gravity of B . The Roche distance of B around the proto-Earth is

$$d_{R,B} \simeq 2.456 \left(\frac{3M_\oplus}{4\pi\rho_B} \right)^{1/3} \approx 2.75 R_\oplus \quad (\rho_B = 3.93 \text{ g cm}^{-3}). \quad (6.21)$$

The value $d_{R,B} \simeq 2.75 R_\oplus$ is a static tidal reference for B . It is not an absolute boundary: the real response depends on the duration of the passage, internal structure, spin and effective cohesion of B .

As a first approximation, for $q_B \gtrsim d_{R,B}$, B crosses the encounter without significant mass loss; for $q_B < d_{R,B}$, a fraction of its matter may be ceded to the disk ($M_{B \rightarrow D} > 0$) and must be followed by the isotopic constraint of Section 14.4. The Level I calculation stops when

$$|\mathbf{a}_{B,\text{diff}}| < a_{\text{tol}} \quad (6.22)$$

over the entire Earth–disk system, that is, when the differential forcing by B becomes dynamically negligible.

6.9 After the passage: return to the heliocentric problem

At the end of the local encounter, B no longer belongs to the Earth–disk system. Its relevant state is no longer its geocentric periapsis, but its post-passage heliocentric orbit $\mathcal{H}_{B,1}$ defined by equation (4.14). The problem is therefore split into two levels:

local level: $\Theta_I \rightarrow \mathcal{D}_1$,	heliocentric level: $\mathcal{H}_{B,1} \rightarrow B \in \mathcal{P}_{\text{elim}}$.
---	---

(6.23)

The first level computes the reorganization of the circumterrestrial reservoir; the second verifies that the orbit left to B can be diffused, destabilized and ultimately eliminated in a global instability.

Quantitative control

The near-Earth passage alone is not sufficient to make B disappear. The disappearance of B must be tested by heliocentric integrations including at least the Sun, the growing planets and the migrating giant planets. The theory therefore requires two closures: a local closure of the Earth–disk– B balance, and then a heliocentric closure showing that B need not survive in the final Solar System.

Chapter 7

Injection of superficial proto-terrestrial matter

The preexisting reservoir supplies most of the matter. This chapter describes the second, more limited channel: stripping, by the tide of B , of superficial matter from low latitudes.

7.1 Role of Earth's rotation

The preexisting disk allows the theory to operate even if the encounter injects only a fraction of new matter. Proto-terrestrial rotation remains central in selecting low-latitude material. The monopolar equatorial effective gravity and the dynamical latitude are

$$g_{\text{eff,eq}} \simeq \frac{GM_{\oplus}}{R_{\text{eq}}^2} (1 - \tilde{\omega}^2), \quad \lambda_{\Omega} = \arcsin(\hat{\mathbf{r}} \cdot \hat{\boldsymbol{\Omega}}_{\oplus}). \quad (7.1)$$

The candidate source is a low-latitude band around the instantaneous rotational equator of the proto-Earth.

7.2 Injection threshold

Equatorial matter is driven toward the disk when the tidal acceleration of B , of order $2GM_B R_{\text{eq}}/q_B^3$, exceeds the effective gravity (7.1). The threshold is

$$1 - \tilde{\omega}^2 \lesssim 2 F(\lambda_B) \frac{M_B}{M_{\oplus}} \left(\frac{R_{\text{eq}}}{q_B} \right)^3 \quad (7.2)$$

where $F(\lambda_B)$ is given by Eq. (6.14). For $M_B = 0.1 M_{\oplus}$, $q_B = 2 R_{\text{eq}}$ and $F = 1$, it gives $\tilde{\omega} > 0.987$; for $M_B = 0.5 M_{\oplus}$, $q_B = 2.5 R_{\text{eq}}$ and $F = 1$, $\tilde{\omega} > 0.967$. This static expression is completed in Phase B by the dynamical treatment of the tide, governed by $\eta_{\oplus}$. For a strongly oblate proto-Earth or one close to breakup, the calculation must use the full potential $\Phi_{\oplus}$.

Physical reading

Injection of proto-terrestrial matter by B is more efficient when the proto-Earth is close to critical rotation: fast rotation and grazing passage act together. The preexisting disk supplies most of the reservoir; injected matter enriches it.

7.3 Transient potential

In the frame centered on and rotating with the proto-Earth, an instantaneous diagnostic potential is

$$\Phi_{\star}(\mathbf{r}, t) = \Phi_{\oplus}(\mathbf{r}) - \frac{GM_B}{|\mathbf{r} - \mathbf{r}_B(t)|} + GM_B \frac{\mathbf{r} \cdot \mathbf{r}_B(t)}{|\mathbf{r}_B(t)|^3} - \frac{1}{2} |\boldsymbol{\Omega}_{\oplus} \times \mathbf{r}|^2, \quad (7.3)$$

whose gradient gives back acceleration (6.9) augmented by the non-inertial terms. The instantaneous saddle points verify

$$\nabla \Phi_{\star}(\mathbf{r}_{\text{col}}, t) = \mathbf{0} \quad (7.4)$$

and locate transient gravitational passages on the near side and opposite side of B .

7.4 Injected matter flux

The materially relevant output and the total injected mass are

$$\frac{dM_{\oplus \rightarrow D}}{d\lambda_{\Omega} dz dt}, \quad M_{\oplus \rightarrow D} = \iiint \frac{dM_{\oplus \rightarrow D}}{d\lambda_{\Omega} dz dt} d\lambda_{\Omega} dz dt, \quad (7.5)$$

where z measures the depth of origin.

Physical idea

The preexistence of the disk transforms the problem: matter separated from the proto-Earth becomes an enrichment flux of the circumterrestrial reservoir, rather than the sole origin of that reservoir.

Chapter 8

The lunar window: the passage as orbital sorting

This chapter formulates the central result of the passage. The circumterrestrial reservoir is already massive before the encounter; the function of B then becomes very precise:

Central formulation

The secondary circumterrestrial disk is already massive before the encounter. The non-collisional passage of B acts as an *orbital sorting mechanism*: its gravitational torque re-distributes the angular momentum of the disk. A fraction of the matter is moved toward bound states whose angular momentum permits circularization beyond the Roche limit; a lower-angular-momentum fraction re-accretes onto the proto-Earth, while overly energetic elements can escape. The Moon accretes from the bound cohort preserved in the satellite window.

8.1 Criterion for membership in the lunar window

A matter element belongs to the lunar window if it simultaneously verifies

$$\varepsilon < 0, \quad q_p > R_{\text{eq}}, \quad r_{\text{circ}} = \frac{j^2}{GM_{\oplus}} \geq a_{\text{Roche}}. \quad (8.1)$$

These three conditions respectively state that matter remains bound to Earth, avoids immediate re-accretion onto the proto-Earth, and has enough angular momentum after circularization to remain in the region where satellite accretion can develop. Here ε is the specific orbital energy, q_p the perigee distance and j the specific angular momentum.

8.2 A geometrical relation between the three conditions

For a bound Keplerian orbit of semi-major axis a and eccentricity e ,

$$r_{\text{circ}} = a(1 - e^2) = q_p(1 + e), \quad q_p \leq r_{\text{circ}} < 2q_p \quad (8.2)$$

(demonstration in Appendix A.4). The condition $r_{\text{circ}} \geq a_{\text{Roche}}$ therefore implies $q_p > a_{\text{Roche}}/2 \approx 1.45 R_{\oplus}$: the non-reaccretion condition is almost automatically satisfied, except for a very flattened proto-Earth. *Angular momentum decides*. For inclined elements, the projection j_z on the final angular momentum of the reservoir is used, because relaxation collisions compensate the perpendicular components: $r_{\text{circ}} = j_z^2/(GM_{\oplus})$.

8.3 Measuring the success of the passage

The quantity that directly measures the success of the passage is the mass fraction placed in the lunar window:

$$\eta_L = \frac{M(\varepsilon < 0, q_p > R_{\text{eq}}, r_{\text{circ}} \geq a_{\text{Roche}})}{M_{D,0}}. \quad (8.3)$$

The other outcomes are measured by

$$\begin{aligned} \eta_{\text{reaccr}} &= \frac{M_{\text{reaccr}}}{M_{D,0}}, & \eta_{\text{esc}} &= \frac{M_{\text{esc}}}{M_{D,0}}, & \eta_{\rightarrow B} &= \frac{M_{\rightarrow B}}{M_{D,0}}, \\ \eta_{\text{in}} &= \frac{M(\varepsilon < 0, q_p > R_{\text{eq}}, r_{\text{circ}} < a_{\text{Roche}})}{M_{D,0}}. \end{aligned} \quad (8.4)$$

The fraction η_{in} forms the inner Roche-interior disk: it is not lost and can later feed lunar accretion by viscous spreading [13]. The partition is complete:

$$\boxed{\eta_L + \eta_{\text{in}} + \eta_{\text{reaccr}} + \eta_{\text{esc}} + \eta_{\rightarrow B} = 1.} \quad (8.5)$$

Part of the reservoir located beyond Roche already satisfies criterion (8.1) before the encounter. The specific contribution of B is therefore

$$\Delta\eta_L = \eta_L - \eta_L^{(0)}, \quad \eta_L^{(0)} = \frac{M(r > a_{\text{Roche}})}{M_{D,0}} \text{ before the encounter,} \quad (8.6)$$

measured by comparison with a control calculation without B .

8.4 Numerical objective

The true numerical objective of the theory is

$$\boxed{(M_B, q_B, v_\infty, i_B, \omega_B, \mathcal{D}_0) \longrightarrow (\eta_L, \Delta\eta_L, \eta_{\text{in}}, \eta_{\text{reaccr}}, \eta_{\text{esc}}, L_{D,1}, M_{D,1}).} \quad (8.7)$$

A successful passage is not the one that perturbs the system most strongly. It is the one that maximizes a window:

$$\boxed{\eta_L \text{ high,} \quad \eta_{\text{esc}} \text{ moderate,} \quad M_{\text{reaccr}} \text{ compatible,}} \quad (8.8)$$

while closing the angular-momentum balance (9.5).

8.5 Anatomy of sorting: the tangential impulse

Consider an element on a circular orbit of radius r_0 that receives a tangential impulse Δv_t . Define $\delta = \Delta v_t / v_K(r_0)$. Its circularization radius becomes

$$r_{\text{circ}} = r_0 (1 + \delta)^2, \quad (8.9)$$

and the three sorting boundaries are (Appendix A.5)

$$\delta_{\text{re}}(r_0) = \sqrt{\frac{2R_{\text{eq}}}{r_0 + R_{\text{eq}}}} - 1, \quad \delta_L(r_0) = \sqrt{\frac{a_{\text{Roche}}}{r_0}} - 1, \quad \delta_{\text{esc}} = \sqrt{2} - 1 \approx 0.414. \quad (8.10)$$

An element re-accretes if $\delta < \delta_{\text{re}}$, joins the inner disk if $\delta_{\text{re}} < \delta < \delta_L$, enters the lunar window if $\delta_L \leq \delta < \delta_{\text{esc}}$, and escapes beyond.

Table 8.1: Orbital sorting boundaries for a tangential impulse ($R_{\text{eq}} = R_\oplus$, $a_{\text{Roche}} = 2.9 R_\oplus$).

$r_0/R_\oplus$	δ_{re} (re-accretion)	δ_L (lunar window)	δ_{esc} (escape)
2.0	-0.18	+0.20	+0.41
2.5	-0.24	+0.08	+0.41
3.0	-0.29	-0.02	+0.41
4.0	-0.37	-0.15	+0.41
5.0	-0.42	-0.24	+0.41

A purely radial impulse does not change j , and therefore does not change r_{circ} : it changes eccentricity and energy and can lead to re-accretion or escape, but it does not move an element into the lunar window. *Sorting toward the window is governed by torque*, that is, by the tangential component of the impulse.

8.6 Tidal impulse: analytical form and order-of-magnitude bound

The impulse received by a reservoir element is given by the closed form of Appendix A.6, valid under the three stated hypotheses: rectilinear trajectory, impulsive forcing, fixed particle. The first-order approximation $\Delta \mathbf{v} \propto (0, y, -z)$ is reliable only for $r \ll q_B$; it loses all precision once $r/q_B \gtrsim 0.3$ and must not serve as reference in the region feeding the lunar window.

To fix orders of magnitude, the bound

$$\frac{\Delta v}{v_K} \sim \sqrt{2} \frac{\mu_B}{\sqrt{1 + \mu_B}} \left(\frac{r}{q_B} \right)^{3/2} \quad (8.11)$$

is used (Appendix A.6). For $\mu_B = 0.1$, the amplitude is about $0.13 (r/q_B)^{3/2}$; for $\mu_B = 0.3$, about $0.37 (r/q_B)^{3/2}$. The sign of the tangential component alternates with the azimuth of the element relative to B : part of the reservoir gains angular momentum, another part loses it. This is literally orbital sorting.

The hyperbolic curvature correction is treated in Phase B by direct integration on the true conic, and the deviation from the closed form (A.4) is an output of the calculation. This is the purpose of gate $\mathcal{D}_{\text{imp}}$ defined below.

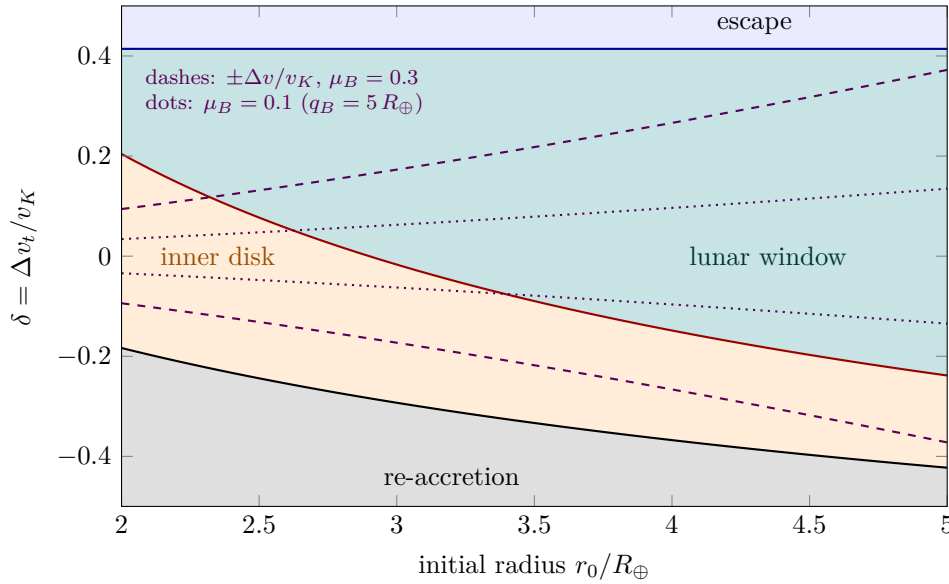


Figure 8.1: Analytical map of orbital sorting for a tangential impulse: re-accretion, Roche-interior disk, lunar window and escape as functions of the initial radius r_0 and relative impulse δ . The violet curves give the order-of-magnitude amplitude of the tidal impulse (8.11) for a passage at $q_B = 5 R_\oplus$. This is an order-of-magnitude map, replaced in Phase B by the exact calculation.

Quantitative control

Figure 8.1 shows that the lunar window is accessible: for $\mu_B \sim 0.3$ and a passage at $q_B = 5 R_\oplus$, matter initially near $2.5\text{--}3 R_\oplus$ receives impulses that make it enter the window, without reaching the escape threshold or the re-accretion threshold. The exact boundaries of this region are an output of Phase B.

8.7 From orbital sorting to a dominant satellite

Orbital sorting also gives a physical explanation for the formation of a dominant satellite. If B excites a reservoir already containing aggregates or moonlets, it can produce

$$\begin{aligned} \text{orbital excitation} &\rightarrow \text{orbit crossings} \rightarrow \text{collisions among circumterrestrial bodies} \\ &\rightarrow \text{coalescence} \rightarrow f_{\text{dom}} \rightarrow 1. \end{aligned} \quad (8.12)$$

The collisions involve the matter of the reservoir and its aggregates; the Earth– B encounter remains non-collisional. N -body simulations of a circumterrestrial disk indeed show that such a disk tends to form a single satellite [31].

Observable consequence

In Phase C simulations, the number of surviving large bodies must decrease after the passage, and the satellite mass must concentrate into one dominant body: $N_{\text{sat}} \rightarrow 1$, $f_{\text{dom}} \rightarrow 1$.

Chapter 9

Balances of the encounter

The encounter ends when the forcing by B becomes negligible. This chapter closes its mass and angular-momentum balances in the language of reservoirs: the reservoir mass is distributed according to partition (8.5), and the total angular momentum (4.6) is conserved.

9.1 Mass balance

The post-encounter disk verifies

$$M_{D,1} = M_{D,0} + M_{\oplus \rightarrow D} + M_{B \rightarrow D} - M_{\text{reaccr}} - M_{\text{esc}} - M_{\rightarrow B}. \quad (9.1)$$

It is decomposed by provenance, each term being net of the losses undergone by the corresponding matter:

$$M_{D,1} = M_{D,1}^{(0)} + M_{D,1}^{(\oplus)} + M_{D,1}^{(B)}, \quad (9.2)$$

where $M_{D,1}^{(0)}$ is the mass of the preexisting disk retained in the reorganized disk, $M_{D,1}^{(\oplus)}$ the injected and retained proto-terrestrial mass, and $M_{D,1}^{(B)}$ the mass possibly ceded by B and retained. This decomposition directly feeds the geochemical fractions of Chapter 13.

9.2 Angular-momentum balance

The balance is written in the barycentric frame of the Earth– B pair, inertial on the encounter time scale (the solar torque is negligible over t_{enc}). The geocentric frame of Eq. (6.9) is not inertial; it is used for trajectory integration, but not for closing the balance. Before and after the encounter,

$$J_{\text{before}} = S_{\oplus,0} + S_{B,0} + L_{D,0} + L_{\text{orb},0}, \quad (9.3)$$

$$J_{\text{after}} = S_{\oplus,1} + S_{B,1} + L_{D,1} + L_{\text{orb},1} + L_{\text{esc}}, \quad (9.4)$$

where L_{orb} is the orbital angular momentum of the relative Earth– B motion and S_B the spin of B , modified by tidal torques. Closure requires

$$J_{\text{before}} = J_{\text{after}} \quad (9.5)$$

and the Earth–disk system left by the passage is characterized by

$$J_{\oplus D,1} = S_{\oplus,1} + L_{D,1}. \quad (9.6)$$

Quantitative control

The angular-momentum transfer toward the trajectory of B is an output of balance closure during the encounter. The central diagnostic is the Earth–disk state produced after the passage, $J_{\oplus D,1}$, to be compared with the angular momentum of the Earth–Moon system.

Didactic reminder

At the end of the encounter, B has ceased to act. The circumterrestrial reservoir has been sorted: one bound cohort occupies the lunar window, another forms the inner disk, and part has re-accreted or escaped. The next part follows the evolution of this reservoir, from its supply before the encounter to lunar accretion.

Part VI

The secondary disk as an evolving reservoir

The secondary disk is not treated here as a fixed object, but as an evolving reservoir $\mathcal{R}_D(t)$: it is fed, transports angular momentum, cools, condenses, forms aggregates, is sorted by the passage of B , then relaxes and accretes the Moon. Its state at each instant combines components of different nature,

$$\mathcal{R}_D(t) = \mathcal{D}_{\text{gas}}(t) \cup \mathcal{D}_{\text{multiphase}}(t) \cup \mathcal{D}_{\text{particulate}}(t) \cup \mathcal{G}(t), \quad (9.7)$$

where $\mathcal{G}(t)$ denotes the population of aggregates and moonlets. The three chapters in this part follow this reservoir before, during and after the passage.

Chapter 10

Feeding and structure of the reservoir before the passage

10.1 Accretional origin of the reservoir

10.1.1 Matter supply in the Hill sphere

The Hill radius of the proto-Earth on its heliocentric orbit is

$$R_H = a_{\oplus} \left(\frac{M_{\oplus}}{3M_{\odot}} \right)^{1/3}. \quad (10.1)$$

Matter entering this region may carry non-zero geocentric angular momentum. Pebble accretion transmits prograde angular momentum [7, 8], and hydrodynamical simulations resolve gas and pebble flows around rocky embryos up to one Earth mass [21]. The general physics of accretion disks then links dissipation and angular-momentum transport [6].

The disk mass balance is

$$\frac{dM_D}{dt} = \dot{M}_{\text{in}} - \dot{M}_{\oplus} - \dot{M}_{\text{out}}, \quad (10.2)$$

where $\dot{M}_{\text{in}}$ is given by Eq. (3.8), $\dot{M}_{\oplus}$ denotes re-accretion onto the proto-Earth, and $\dot{M}_{\text{out}}$ losses toward the exterior.

10.1.2 Mass and surface density

The instantaneous disk mass and its ratio to the proto-terrestrial mass are

$$M_D(t) = 2\pi \int_{R_{\text{in}}(t)}^{R_{\text{out}}(t)} \Sigma(r, t) r dr, \quad \mu_D(t) = \frac{M_D(t)}{M_{\oplus}}. \quad (10.3)$$

Physical reading

M_D is an output of the feeding and loss balance. The ratio of about 10^{-4} obtained for giant planets in continuously fed models [10] is an astrophysical reference; it does not fix μ_D for the proto-Earth, whose feeding mode is different (Section 3.3).

10.1.3 Angular momentum supplied by accretion

For a closed control surface $\mathcal{S}_H$ surrounding the terrestrial region of influence, with outward normal $\mathbf{n}$, the incoming angular-momentum flux is

$$\dot{L}_{\text{in}} = - \oint_{\mathcal{S}_H} \rho (\mathbf{r} \times \mathbf{v})_z (\mathbf{v} \cdot \mathbf{n}) d\mathcal{S}, \quad (10.4)$$

the minus sign making an incoming contribution positive. The disk angular-momentum balance is

$$\frac{dL_D}{dt} = \dot{L}_{\text{in}} - \dot{L}_{\oplus} - \dot{L}_{\text{out}} + \mathcal{T}_{\oplus D} + \mathcal{T}_{\odot D}, \quad (10.5)$$

with $\mathcal{T}_{\oplus D}$ and $\mathcal{T}_{\odot D}$ the torques exerted by the proto-Earth and the Sun.

10.1.4 Origin of proto-terrestrial spin

Pebble accretion can transmit prograde angular momentum to the central body [7, 8]. The rotation of the proto-Earth therefore belongs to the same accretion problem as the disk. The reduced rotation is defined as

$$\tilde{\omega} = \frac{\Omega_{\oplus}}{\sqrt{GM_{\oplus}/R_{\text{eq}}^3}}. \quad (10.6)$$

For an approximately spin-conserving contraction,

$$S_{\oplus} \simeq k M_{\oplus} R_{\text{eq}}^2 \Omega_{\oplus} \simeq \text{constant} \implies \tilde{\omega} \propto R_{\text{eq}}^{-1/2} \quad (10.7)$$

when k and $M_{\oplus}$ vary little: thermal contraction of a hot proto-Earth brings it closer to critical rotation.

Physical idea

The disk and Earth's spin are not two independent reservoirs added together. Both result from how matter and angular momentum are distributed during the growth of the proto-Earth; the channel $\dot{M}_{\text{rot}}$ directly links them.

10.2 Radial structure

10.2.1 Inner radius

The inner radius corresponds to the transition between the proto-Earth or its dense envelope and a truly orbital regime:

$$R_{\text{in}} \simeq R_{\text{tr}}. \quad (10.8)$$

This transition depends on pressure, rotation, temperature and, if a significant magnetic field is present, on magnetohydrodynamic constraints. For a proto-Earth close to critical rotation, R_{tr} tends toward R_{eq} .

10.2.2 Outer radius

The outer radius is parameterized by

$$R_{\text{out}} = \xi_H R_H, \quad 0 < \xi_H < 1. \quad (10.9)$$

For circumplanetary disks of giant planets, truncation by solar torques leads to $\xi_H \approx 0.3\text{--}0.4$ [24, 25]; in the present theory, ξ_H is an output of heliocentric dynamics and feeding conditions. The bulk of the mass of a solid-fed disk nevertheless concentrates well inside this edge, near the circularization radius of captured matter.

10.2.3 Roche limit and synchronous orbit

For a reference satellite density ρ_s , the fluid Roche limit is

$$a_{\text{Roche}} \simeq 2.456 \left(\frac{3M_{\oplus}}{4\pi\rho_s} \right)^{1/3} \approx 2.9 R_{\oplus} \quad (\rho_s = 3.34 \text{ g cm}^{-3}). \quad (10.10)$$

Its role is distinct from that of R_{in} :

$$R_{\text{in}} \neq a_{\text{Roche}}. \quad (10.11)$$

A second radius plays an essential dynamical role, the synchronous orbit

$$a_{\text{sync}} = \left(\frac{GM_{\oplus}}{\Omega_{\oplus}^2} \right)^{1/3} \approx 2.0\text{--}2.3 R_{\oplus} \quad (P_{\oplus} = 4\text{--}5 \text{ h}). \quad (10.12)$$

For a rapidly rotating proto-Earth, $a_{\text{sync}} < a_{\text{Roche}}$: any body assembled beyond the Roche limit migrates outward under the effect of terrestrial tides.

Physical reading

A disk can occupy the region interior to the Roche limit. This limit controls the survival and assembly of large self-gravitating aggregates; it is central to lunar accretion, not to the existence of the disk. The ordering $a_{\text{sync}} < a_{\text{Roche}}$ ensures that bodies formed beyond Roche move away from Earth rather than fall back onto it.

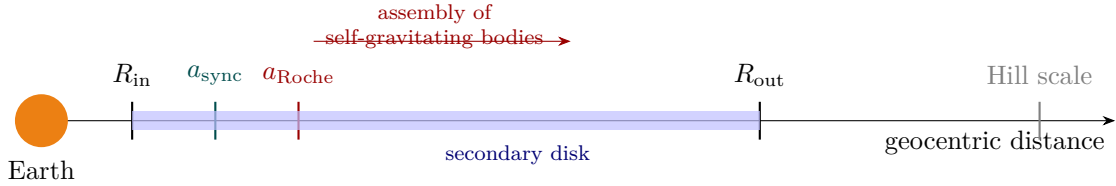


Figure 10.1: Conceptual radial organization of the circumterrestrial disk. The inner, synchronous, Roche, outer and Hill radii play different physical roles (not to scale).

10.3 The reservoir as an accretion disk

Before entering the particular structure of $\mathcal{R}_D$, it is useful to recall that an accretion disk is a *generic* structure: matter in orbit around a central body, viscous dissipation converting orbital energy into heat, and outward angular-momentum transport allowing inward accretion. The three ingredients are inseparable. This is exactly the structure proposed for $\mathcal{R}_D$, and the tools developed for accretion disks in general [4, 6, 5] therefore apply to it without special transposition.

An analogue for the feeding. Disk formation by mass transfer through the inner Lagrange point of a binary system is a known and robust mechanism [5]. The body filling its Roche lobe yields matter to its companion, and that matter, too rich in angular momentum to fall directly, circularizes into a disk. We propose to see in the flux $\Phi_{\oplus \rightarrow D}$ the analogue of this mechanism: the proto-Earth, brought near its rotation limit and deformed by the differential field of B , yields surface material that circularizes in turn. The analogy is not a proof, and the geometries differ; it indicates only that the type of flux invoked is not exotic.

The central problem is angular-momentum transport. In any accretion disk, the angular momentum lost by matter falling toward the centre must be compensated by a gain of angular momentum by matter located farther out. Without this transfer, nothing accretes. This is precisely the function attributed to the orbital sorting of Chapter 8: the torque exerted by B moves angular momentum toward the outside of the reservoir, allowing a fraction of the matter to cross a_{Roche} while another fraction falls back inward. The passage therefore plays, once and impulsively, the role that viscosity plays slowly and continuously.

An additional transport route to consider. The effective viscosity $\nu \sim G^2 \Sigma^2 / \Omega_K^3$ adopted above assumes transport by gravitational torques. If the reservoir retains residual gas or an ionized fraction, the magnetorotational instability [3] becomes a highly efficient mechanism of angular-momentum redistribution in weakly magnetized disks, and could complement or replace this effective viscosity. This route is not adopted here as a baseline hypothesis, because the degree of ionization of $\mathcal{R}_D$ is unconstrained; we propose to treat it as an alternative branch of the viable domain, to be activated if the thermal state computed in Phase A indicates sufficient ionization.

10.4 Orbital structure and angular-momentum transport

10.4.1 Quasi-Keplerian rotation

When the disk mass remains small compared with that of the proto-Earth and pressure provides only a moderate correction, the dominant orbital frequency is

$$\Omega_K(r) = \sqrt{\frac{GM_\oplus}{r^3}}. \quad (10.13)$$

The actual azimuthal velocity is

$$v_\phi^2 = \frac{GM_\oplus}{r} + \frac{r}{\rho} \frac{\partial P}{\partial r} + r a_{\text{sg}}(r) + r a_{J_2}(r), \quad (10.14)$$

where a_{sg} represents the disk self-gravity and a_{J_2} the contribution of the flattening of a rapidly rotating proto-Earth. The specific angular momentum is $j(r) = r v_\phi(r)$.

10.4.2 Global disk angular momentum

In the Keplerian limit,

$$L_D = 2\pi \int_{R_{\text{in}}}^{R_{\text{out}}} \Sigma(r) \sqrt{GM_\oplus r} r dr, \quad (10.15)$$

and the initial Earth–disk system has

$$J_{\oplus D,0} = S_{\oplus,0} + L_{D,0}. \quad (10.16)$$

Physical idea

The angular momentum destined for the future Moon may already be partly orbital before the passage of B . The encounter therefore does not have to create the entire orbital budget: it redistributes a budget already present and adds or removes the part exchanged with the trajectory of B .

10.4.3 Viscous evolution

In the approximation of a thin, axisymmetric, quasi-Keplerian disk, the surface density follows the classical equation [6]

$$\frac{\partial \Sigma}{\partial t} = \frac{3}{r} \frac{\partial}{\partial r} \left[r^{1/2} \frac{\partial}{\partial r} \left(\nu \Sigma r^{1/2} \right) \right]. \quad (10.17)$$

For a gaseous component, $\nu = \alpha c_s H$ with $H \simeq c_s / \Omega_K$. For a dense particulate component, the effective viscosity is dominated by self-gravity wakes and scales as $G^2 \Sigma^2 / \Omega_K^3$ [28]. The characteristic viscous time is

$$t_\nu(r) \sim \frac{r^2}{\nu}. \quad (10.18)$$

Chapter 11

Thermodynamics, stability and persistence

11.1 Thermal and multiphase state

11.1.1 Energy balance

The local energy balance of the disk is

$$Q_{\text{visc}}^+ + Q_{\text{acc}}^+ + Q_{\oplus}^+ + Q_{\odot}^+ = Q_{\text{rad}}^- + Q_{\text{adv}}^- + Q_{\text{phase}}^-. \quad (11.1)$$

In a quasi-Keplerian disk, viscous dissipation per unit surface and radiative cooling from both faces are

$$Q_{\text{visc}}^+ \simeq \frac{9}{4} \nu \Sigma \Omega_K^2, \quad Q_{\text{rad}}^- \simeq 2 \sigma_{\text{SB}} T_{\text{eff}}^4. \quad (11.2)$$

With vertical optical depth $\tau \simeq \kappa \Sigma / 2$, the diffusion approximation gives, for an optically thick medium,

$$T_{\text{mid}}^4 \simeq \frac{3}{4} \tau T_{\text{eff}}^4 + T_{\text{irr}}^4. \quad (11.3)$$

Physical reading

Temperature enters simultaneously into thickness, viscosity, opacity, sound speed and phase state. Thermodynamics directly participates in the orbital structure and lifetime of the disk; it is not a module added after the dynamics.

11.1.2 Condensation and phase transitions

For each species i , the phase state is a function

$$\mathcal{P}_i = \mathcal{P}_i[T(r, t), P(r, t), f_{\text{O}_2}(r, t), \mathbf{X}(r, t)], \quad (11.4)$$

and the solid-evolution chain is

$$\text{vapor} \rightarrow \text{condensates} \rightarrow \text{grains} \rightarrow \text{aggregates} \rightarrow \text{satellitesimals}. \quad (11.5)$$

For a population of particles with number density n_p , collision cross-section σ_c and relative velocity Δv , the collision time has the scale

$$t_{\text{coll}} \sim \frac{1}{n_p \sigma_c \Delta v}. \quad (11.6)$$

11.1.3 Vertical settling

A particle coupled to the gas is characterized by its Stokes number $\text{St} = t_{\text{stop}} \Omega_K$. The competition between settling and turbulence fixes its vertical thickness:

$$\frac{H_p}{H} \sim \left(\frac{\alpha}{\alpha + \text{St}} \right)^{1/2}. \quad (11.7)$$

This relation follows the emergence of a solid-enriched midplane.

11.2 Gravitational stability and persistence

11.2.1 Toomre criterion

The local stability of a self-gravitating thin disk is measured by the Toomre parameter, whose constant depends on the nature of the medium:

$$Q_T^{\text{gas}} = \frac{c_s \kappa_{\text{epi}}}{\pi G \Sigma} \quad [29, 30], \quad Q_T^{\text{part}} = \frac{\sigma_r \kappa_{\text{epi}}}{3.36 G \Sigma} \quad [11], \quad (11.8)$$

where κ_{epi} is the epicyclic frequency and σ_r the radial velocity dispersion of the particles. A cold and dense disk approaches the regime $Q_T \sim 1$.

It is instructive to evaluate this criterion on a reservoir of a few lunar masses. For $M_{D,0} = 6.9 M_L$ distributed as $\Sigma \propto r^{-1}$ between 1.6 and $2.9 R_{\oplus}$, the surface density is $\Sigma \simeq 9.5 \times 10^8 \text{ kg m}^{-2}$ at the inner edge and $5.3 \times 10^8 \text{ kg m}^{-2}$ at the outer edge, and stability $Q_T^{\text{part}} \geq 1$ requires

$$\sigma_r \gtrsim 0.07\text{--}0.10 v_K \quad (390\text{--}470 \text{ m s}^{-1}). \quad (11.9)$$

A reservoir of this density is therefore not a quiescent dust disk: either it is hot, fluid or partially molten, or it is already organized into moonlets. The Phase A calculation decides between these two states, and that choice determines the nature of the Phase B equations, since a fluid reservoir calls for hydrodynamics and a reservoir of moonlets for an N -body treatment. The lighter the reservoir, the lower the required σ_r and the more legitimate the independent particle regime becomes: one more reason to sweep $M_{D,0}$.

Physical idea

The preexistence of the disk raises a decisive question: when B arrives, is the disk still a diffuse reservoir, or has it already formed aggregates and moonlets? The answer can be calculated from thermodynamics, gravitational stability and growth times.

11.2.2 Disk persistence

Persistence is expressed through the state of the reservoir at t_B rather than through an arbitrary duration:

$$M_D(t_B) + \sum_k m_k(t_B) \geq M_{D,\text{min}}, \quad L_D(t_B) + \sum_k L_k(t_B) \geq L_{D,\text{min}}, \quad R_{\text{out}}(t_B) > R_{\text{in}}(t_B). \quad (11.10)$$

To this global condition is added a local one, comparing the reservoir lifetime with the delay actually available before the encounter:

$$t_{\text{pers}}(r) = \min(t_{\nu}(r), t_{\text{frag}}(r), t_{\text{reaccr}}(r), t_{\text{diff}}(r)) > t_B - t_{D,0}. \quad (11.11)$$

The four times are respectively viscous spreading (5.10), gravitational fragmentation associated with Q_T , re-accretion onto the proto-Earth and reservoir diffusion. The condition must hold over the whole dense radial interval, not merely on average.

Falsification criterion

If feeding, cooling and transport laws systematically dissipate or re-accrete the reservoir before any time window compatible with the passage of B , the corresponding branch of the theory is excluded.

The three chronological regimes describing the state of the reservoir at the time of the passage were presented in Chapter 5 (Section 5.5).

Chapter 12

Reservoir reorganization and lunar accretion

After orbital sorting, the reservoir enters a phase of dissipative relaxation, then accretion. This chapter describes the new state of the reservoir, its dissipation, the mass available beyond Roche, and the control of the accessible satellite mass.

12.1 New state of the disk

The passage provides

$$\mathcal{D}_1 = \{M_{D,1}, \Sigma_1, T_1, \mathbf{X}_1, L_{D,1}, e(r), i(r), \mathbf{f}_{\text{phase},1}, \mathcal{G}_1\}, \quad (12.1)$$

and the collective evolution follows

$$\mathcal{D}_1 \longrightarrow \mathcal{D}_{\text{relax}} \longrightarrow \mathcal{D}_{\text{accr}} \longrightarrow \text{proto-Moon}. \quad (12.2)$$

12.2 Dissipation

The relaxation energy balance is

$$E_{\text{diss}} = E_{\text{shock}} + E_{\text{circ}} + E_{\text{vis}} + E_{\text{coll}} - E_{\text{rad}} - E_{\text{esc}}. \quad (12.3)$$

A parcel on an orbit with semi-major axis a and eccentricity e , circularized at conserved angular momentum, reaches the radius $a_{\text{circ}} = a(1 - e^2)$. The specific energy released, positive, is

$$\Delta \varepsilon_{\text{circ}} = \frac{GM_{\oplus}}{2a_{\text{circ}}} - \frac{GM_{\oplus}}{2a} = \frac{GM_{\oplus}}{2a} \frac{e^2}{1 - e^2} \geq 0. \quad (12.4)$$

12.3 Mass available beyond Roche

The mass and angular momentum located beyond the Roche limit are

$$M_{>R} = 2\pi \int_{a_{\text{Roche}}}^{R_{\text{out}}} \Sigma(r) r \, dr, \quad L_{>R} = 2\pi \int_{a_{\text{Roche}}}^{R_{\text{out}}} \Sigma(r) \sqrt{GM_{\oplus} r} \, r \, dr. \quad (12.5)$$

12.4 Disk–satellite accretion control

Once the reorganized disk is established, the normalized specific angular momentum is defined as

$$\ell_D = \frac{L_D}{M_D \sqrt{GM_{\oplus} a_{\text{Roche}}}}. \quad (12.6)$$

The reference disk–satellite accretion relation [12] is

$$\frac{M_{L,\text{Ida}}}{M_D} \simeq 1.9 \ell_D - 1.1 - 1.9 \frac{M_{\text{esc}}}{M_D}. \quad (12.7)$$

This relation, calibrated on particulate disks in which the Moon forms near $1.3 a_{\text{Roche}}$, is used as a first-order estimator. When the disk includes a Roche-interior fluid component, the Moon forms farther away, near $2.15 a_{\text{Roche}}$, and the revised estimate of Salmon and Canup [13] must be used. The computed value is bounded to the physical interval $0 \leq M_{L,\text{Ida}} \leq M_D$.

Physical reading

The Ida relation intervenes after definition of the post-encounter disk. It does not replace hydrodynamics; it gives a rapid control on the mass scale accessible from the pair (M_D, L_D) produced by the passage of B .

Quantitative control

This control is most useful as an upper bound on $M_{D,0}$, and it guides the sweep. For a reservoir of $6.9 M_L$ distributed as $\Sigma \propto r^{-1}$ between 1.6 and $2.9 R_\oplus$ in Keplerian rotation, one obtains $L_{D,0} \simeq 3.8 \times 10^{34} \text{ kg m}^2 \text{ s}^{-1}$, hence $\ell_D \simeq 0.88$ and

$$M_{L,\text{Ida}} \simeq 0.57 M_D \simeq 3.9 M_L. \quad (12.8)$$

The estimator therefore returns a satellite mass larger than the observed one, indicating that the massive branch of the reservoir is not the branch to explore first. Read the other way, the same estimator identifies the useful interval: $M_{D,0} \simeq 2\text{--}3 M_L$ brings $M_{L,\text{Ida}}$ close to $M_{L,\text{obs}}$. This interval should anchor the grid of Table 15.1, the missing mass then being supplied by the injection channel $M_{\oplus \rightarrow D}$ (Section 16.5) rather than by the reservoir alone.

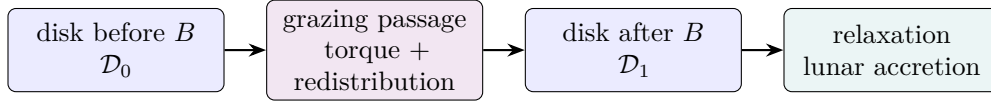


Figure 12.1: The passage of B transforms an existing disk; lunar accretion then acts on the reorganized disk.

Didactic reminder

The dynamical path is now complete: feeding, orbital sorting, relaxation and accretion. The next part verifies that it is also a *material* path: the Moon must preserve the chemical memory of the reservoirs from which it came.

Part VII

Geochemistry: families of tracers

Geochemistry belongs to the same story as dynamics. Each matter element preserves, throughout the path, its dynamical, thermal and chemical labels. This part first sets the framework of the three material lineages, then organizes tests by tracer families, each of which answers a precise physical question.

Chapter 13

Three material lineages

13.1 Fractions of provenance

Lunar matter may contain three contributions:

$$f_0 + f_{\oplus} + f_B = 1, \quad (13.1)$$

where f_0 is the fraction inherited from the preexisting disk, $f_{\oplus}$ the fraction injected from the proto-Earth during the encounter, and f_B the possible contribution from B . The corresponding masses are

$$M_L^{(k)} = f_k M_L, \quad k \in \{0, \oplus, B\}, \quad (13.2)$$

and the f_k follow from decomposition (9.2), weighted by the accretion efficiency of each lineage.

13.2 Concentration-weighted isotopic mixing

For an isotopic system or ratio R_X , mixing is weighted by concentrations:

$$R_X^L = \frac{f_0 C_{X,0} R_{X,0} + f_{\oplus} C_{X,\oplus} R_{X,\oplus} + f_B C_{X,B} R_{X,B}}{f_0 C_{X,0} + f_{\oplus} C_{X,\oplus} + f_B C_{X,B}}. \quad (13.3)$$

This relation applies separately to the O, Ti, Cr, Si and W systems, each with its own concentrations.

Physical idea

The preexisting disk has its own chemical history; its composition cannot be made invisible by definition. The theory draws a prediction from this: lunar chemistry results from a mixture whose fractions are computed by dynamics.

13.3 One physical question per tracer family

Geochemical tracers are not interchangeable: each records a different process. The theory groups them into four families (Figure 13.1).

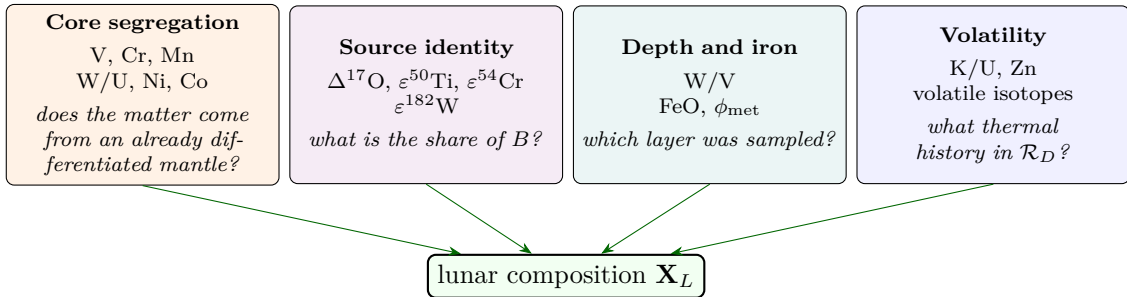


Figure 13.1: The four tracer families and the physical question answered by each.

Chapter 14

Families of tracers

14.1 Moderately siderophile refractory elements: V, Cr, Mn

Chromium, vanadium and manganese have similar abundances in Earth's mantle and in the Moon, and they are clearly depleted relative to their primitive abundances normalized to magnesium [38]. This similarity has been used to suggest that the Moon derives substantially from Earth's mantle after core formation [39].

Vanadium has a particular place in this family. It is less volatile than magnesium: its depletion cannot result from evaporative loss, either in the nebula or in the circumterrestrial reservoir. It records metal–silicate segregation.

The debate must be reported faithfully. Experiments at 1 bar showed that the depletions of V, Cr and Mn do not follow metal/silicate partition coefficients measured under those conditions, which led to the conclusion that they were not primarily explained by terrestrial core formation [39]. High-pressure experiments subsequently showed that, under lower-mantle conditions, the behavior of these elements changes with the increasing solubility of oxygen in liquid iron, so that similar depletion patterns in Earth and Moon indicate that a large fraction of the Moon comes from Earth's mantle after core segregation [38, 40].

Physical reading

A small Moon at low internal pressure hardly produces by itself a vanadium depletion characteristic of high-pressure segregation. In the present theory, this signature is interpreted as the trace of a flux of terrestrial silicate matter toward the reservoir, $\Phi_{\oplus \rightarrow D}$. Vanadium is not read in isolation: it is interpreted together with tungsten and the other tracers of the same lineage.

14.2 Tungsten: abundance, isotopes and clock

Tungsten provides two independent pieces of information.

Its abundance. The W/U, or W/Th, ratio compares elements of similar incompatibility during silicate melting. It is therefore only weakly sensitive to magmatic differentiation and records the siderophile behavior of tungsten, that is, metal segregation [41]. A lunar value close to the terrestrial value indicates a silicate source that underwent the same segregation history.

Its isotopes. After correction for late accretion, lunar $\epsilon^{182}\text{W}$ is almost identical to that of Earth's mantle [34, 35]. In the present theory, this identity corresponds to what is expected, since lunar tungsten comes mainly from terrestrial silicate matter; this expectation must be checked in the viable domain.

Original contribution of the theory

Two results follow here from the very structure of the proposed path, with no adjusted parameter.

First the Earth–Moon isotopic similarity. If lunar matter comes mainly from the terrestrial mantle, through the preexisting reservoir and through the flux $\Phi_{\oplus \rightarrow D}$, and if the contribution of B remains small, then identity in O, Ti, Cr and ^{182}W is not a coincidence to be explained: it is the direct consequence of provenance. The theory does not make this identity *possible*, it

makes it *expected*.

Second the V, Cr and Mn depletions. These elements are moderately siderophile, and their deficit in the terrestrial mantle is inherited from core segregation. If the flux $\Phi_{\oplus \rightarrow D}$ draws terrestrial silicate material already depleted, the Moon inherits this signature without any need to invoke a second lunar segregation. The same lineage of matter therefore carries at once the mass, the metallic-iron deficit and the moderately siderophile element ratios. It is this economy of hypotheses that we propose as the main argument in favour of the path studied.

A clock. If the sampled layer had acquired an Hf/W ratio different from that of the mean mantle, decay of ^{182}Hf (half-life 8.9 Ma) would have produced an excess or deficit of ^{182}W , unless ^{182}Hf was already extinct. Tungsten therefore constrains both the source and the time t_B .

Bound on reservoir composition. With $f_B = 0$ and the silicate Earth as reference ($\varepsilon^{182}\text{W}_{\oplus} = 0$), the linearized form of Eq. (13.3) gives (Appendix A.9)

$$\varepsilon^{182}\text{W}_L \simeq f_0 \frac{C_{\text{W},0}}{C_{\text{W},\oplus}} \varepsilon^{182}\text{W}_0 \implies f_0 \lesssim \frac{\delta\varepsilon C_{\text{W},\oplus}}{C_{\text{W},0} |\varepsilon^{182}\text{W}_0|}, \quad (14.1)$$

where $\delta\varepsilon$ is the tolerated Earth–Moon difference; this bound applies if the reservoir material is chondritic. With $C_{\text{W},0} \approx 0.1\text{--}0.2$ ppm, $\varepsilon^{182}\text{W}_0 \approx -1.9$, $C_{\text{W},\oplus} \approx 0.013$ ppm and $\delta\varepsilon \approx 0.1\text{--}0.2$, chondritic matter could represent only 0.3 to 1.4 % of the Moon.

Physical reading

Because the theory assigns the principal share of lunar matter to the preexisting reservoir, bound (14.1) requires this reservoir to be of *terrestrial silicate* nature: $\varepsilon^{182}\text{W}_0 \simeq 0$, and the bound disappears. This points to equatorial mass loss $\Phi_{\oplus \rightarrow D}^{\text{rot}}$ and ordinary mantle-impact ejecta as the dominant supply of $\mathcal{R}_D$, with co-accretion of chondritic matter remaining minor. This is a prediction of the theory: the composition of the circumterrestrial reservoir is terrestrial.

14.3 Iron and siderophiles: FeO, Ni, Co and metal

Most refractory lithophile and weakly siderophile elements have similar abundances in the Moon and silicate Earth, but FeO appears significantly higher in the lunar mantle, which is an important constraint to be satisfied [42]. The framework of three lineages makes it possible to treat this quantitatively.

14.3.1 FeO and silicate reservoir

For a simplified mixture, the FeO content of the disk after the encounter is

$$X_{\text{FeO}}^{D,1} = f_0 X_{\text{FeO}}^0 + f_{\oplus} X_{\text{FeO}}^{\oplus} + f_B X_{\text{FeO}}^B, \quad (14.2)$$

then corrected for fractionations due to melting, condensation and vapor loss. If a melt fraction F is extracted from a reservoir with a bulk partition coefficient D_X , a fractional-melting law for example gives

$$C_X^{\text{liq}} = C_{X,0} \frac{1 - (1 - F)^{1/D_X}}{F}. \quad (14.3)$$

The calculation does not choose one depth for FeO, another for W and a third for isotopes: the same material lineage carries all properties.

14.3.2 Metal and silicate

The disk is decomposed into

$$M_D = M_D^{\text{sil}} + M_D^{\text{met}} + M_D^{\text{vap}}, \quad \phi_{\text{met}} = \frac{M_D^{\text{met}}}{M_D^{\text{sil}} + M_D^{\text{met}}}. \quad (14.4)$$

The siderophile elements Ni, Co and W constrain the origin, segregation and re-accretion of metal.

Observable consequence

The lunar FeO excess must be reproduced by the calculation of mixing and internal transformations within the reservoir (Π_D in Eq. (4.8)): oxidation of part of the metal, fractionation during condensation, or a more oxidized silicate component. Because bound (14.1) limits the chondritic share, the excess must come from internal reservoir processes.

14.4 O, Ti, Cr isotopes: the imprint of B

The oxygen, titanium and chromium isotopic systems distinguish Solar System bodies according to their formation reservoir. They therefore measure the share of B in the Moon. For oxygen, at comparable concentrations,

$$f_B \lesssim \frac{\delta_{17}}{|\Delta^{17}\text{O}_B - \Delta^{17}\text{O}_\oplus|}, \quad (14.5)$$

where δ_{17} , of order a few ppm, is the measured Earth–Moon difference [36]. For a body B with Martian-like isotopic composition ($|\Delta^{17}\text{O}_B - \Delta^{17}\text{O}_\oplus| \approx 300$ ppm), $f_B \lesssim 2\%$. Titanium [37] and chromium constraints are expressed in the same way.

Original contribution of the theory

The mechanism naturally favors a small material contribution from B , because the passage is non-collisional. The Earth–Moon isotopic similarity therefore becomes an expected constraint of the viable domain, to be checked through the bounds on f_B (Eq. (14.5)).

14.5 Volatiles

For a volatile species i , the balance and isotopic fractionation associated with vapor loss of Rayleigh type are

$$\frac{dM_i}{dt} = \dot{M}_{i,\text{in}} - \dot{M}_{i,\oplus} - \dot{M}_{i,\text{esc}} + \dot{M}_{i,\text{cond}} - \dot{M}_{i,\text{vap}}, \quad R_i = R_{i,0} f_i^{(\alpha_i-1)}, \quad (14.6)$$

where f_i is the residual fraction and α_i the effective fractionation factor.

14.6 The W/V ratio and depth of origin

Tungsten and vanadium are not isolated proofs: they are *coupled tracers* of fluxes between reservoirs and of the depth of the sampled matter. Their interpretation passes through the composition equation (4.8) and the injected-matter distribution (7.5).

Tungsten is highly incompatible: during crystallization of a magma ocean, it concentrates in the last liquids and therefore toward the surface. Vanadium is moderately compatible, retained by pyroxenes and spinels. The W/V ratio, normalized to reference elements with similar behavior, therefore varies with the depth and differentiation state of the sampled layer. For injected matter it is

$$\left(\frac{\text{W}}{\text{V}}\right)_{\oplus \rightarrow D} = \frac{\iiint \frac{dM_{\oplus \rightarrow D}}{d\lambda_\Omega dz dt} C_W(\lambda_\Omega, z) d\lambda_\Omega dz dt}{\iiint \frac{dM_{\oplus \rightarrow D}}{d\lambda_\Omega dz dt} C_V(\lambda_\Omega, z) d\lambda_\Omega dz dt}. \quad (14.7)$$

Physical idea

This is the most direct expression of the single material lineage: the same depth distribution z and latitude distribution λ_Ω , produced by the passage dynamics (Eq. (7.5)), simultaneously fix the injected mass and its elemental ratios. One does not choose a separate depth for each tracer.

14.7 Synthesis of tracers

Table 14.1: Tracer families, recorded processes and predictions of the theory.

Tracer	What it records	Prediction of the theory
V, Cr, Mn	high-pressure core segregation	lunar depletions similar to terrestrial ones
W/U, W/Th $\varepsilon^{182}\text{W}$ $\Delta^{17}\text{O}$, $\varepsilon^{50}\text{Ti}$, $\varepsilon^{54}\text{Cr}$	siderophile behavior of tungsten source identity, chronology share of B	lunar ratio close to the terrestrial ratio close to Earth's mantle; constraint on t_B $f_B \ll 1$
FeO, ϕ_{met} W/V	oxidized iron and metal depth and differentiation of the sampled layer	lunar excess produced in the reservoir linked to stripped latitude and depth
K/U, Zn	thermal history of the reservoir	depletion produced in $\mathcal{R}_D$

Didactic reminder

Dynamics and geochemistry now describe the same path. The next part gathers the conditions that it must satisfy simultaneously: the viability domain.

Part VIII

Viability domain

A physical path is credible only if it exists within a continuous region of parameter space, not for a single finely tuned configuration. This part gathers the Level I parameters, the gates that the path must pass simultaneously, and the numerical program that will produce the viability maps.

Chapter 15

Parameters, gates and viable domain

15.1 Level I parameters

The vector Θ_I (Eq. (4.13)) groups the parameters of the proto-Earth, the reservoir and the encounter. Table 15.1 proposes an initial exploration grid; the prograde low-latitude band is designated as the priority region.

Table 15.1: Initial exploration grid for Level I parameters.

Parameter	Initial domain	Comment
$\mu_B = M_B/M_\oplus$	0.05–1	mass of B
q_B	$\chi_{\text{tide}} d_{R,B} - 10 R_\oplus$	lower bound: gate (15.6), i.e. $\approx 3.0 R_\oplus$
v_∞	0–10 km s ^{−1}	relative asymptotic velocity
t_{ej}	$> t_B$	later elimination or ejection of B
i_B	0–180°	priority: $ \lambda_B \leq \lambda_c$, prograde
ω_B	0–180°	priority: near 90°
$\tilde{\omega}$	0.8–1	reduced proto-Earth rotation
$R_{\text{eq}}/R_\oplus$	1–1.3	flattening
Reservoir	2–5 $R_\oplus$, $\Sigma \propto r^{-p}$, $p \in [0, 3]$	radial structure
$M_{D,0}$	1–8 M_L	initial reservoir mass; priority 2–3 M_L
$\Omega_\oplus$	$\tilde{\omega} \in [0.8, 1]$	controls $M_{\oplus \rightarrow D}$ (Section 16.5)

15.2 Gates of the viable domain

The total domain is defined by the intersection

$$\mathcal{D}_{\text{viable}} = \mathcal{D}_{\text{disk},0} \cap \mathcal{D}_{\text{thermo}} \cap \mathcal{D}_{\text{grav}} \cap \mathcal{D}_{\text{nc}} \cap \mathcal{D}_{\text{nt}} \cap \mathcal{D}_B \cap \mathcal{D}_{B,\text{hel}} \cap \mathcal{D}_{\text{tide},B} \cap \mathcal{D}_{\text{fen}} \cap \mathcal{D}_{\text{imp}} \cap \mathcal{D}_J^{\text{num}} \cap \mathcal{D}_J^{\text{EM}} \cap \mathcal{D}_{\text{accr}} \cap \mathcal{D}_{\text{arch}} \cap \mathcal{D}_{\text{geo}} \cap \mathcal{D}_{\text{iso}} \cap \mathcal{D}_{\text{chrono}}. \quad (15.1)$$

The gates have the following meanings:

$$\mathcal{D}_{\text{disk},0} = \{\Theta_I : \text{conditions (11.10) and (11.11) verified at } t_B\}, \quad (15.2)$$

$$\mathcal{D}_{\text{thermo}} = \left\{ \Theta_I : f_{\text{sol}}(r, t) > 0 \text{ for } r > a_{\text{Roche}} \text{ before } t_{\text{acc},L}^{\text{start}} \right\}, \quad (15.3)$$

$$\mathcal{D}_{\text{grav}} = \{\Theta_I : Q_T(r, t) \text{ and } t_{\text{sat}} \text{ compatible with the regime of Section 5.5}\}, \quad (15.4)$$

$$\mathcal{D}_{\text{nc}} = \{\Theta_I : q_B > R_{\text{eq}} + R_{B,\text{eff}}\}, \quad (15.5)$$

$$\mathcal{D}_{\text{tide},B} = \{\Theta_I : q_B > \chi_{\text{tide}} d_{R,B}, \chi_{\text{tide}} > 1\}, \quad (15.6)$$

$$\mathcal{D}_{\text{nt}} = \left\{ \Theta_I : R_{\text{out,dense}} < \min(r_{\text{node}}^+, r_{\text{node}}^-) \right\}, \quad (15.7)$$

$$\mathcal{D}_B = \left\{ \Theta_I : f_B = \frac{M_{B \rightarrow L}}{M_L} \leq f_{B,\text{max}} \right\}, \quad (15.8)$$

$$\mathcal{D}_{B,\text{hel}} = \left\{ \Theta_I^+ : t_B < t_{\text{ej}} \text{ and } \mathcal{H}_{B,1} \rightarrow B \in \mathcal{P}_{\text{elim}} \right\}, \quad (15.9)$$

$$\mathcal{D}_{\text{fen}} = \{\Theta_I : \Delta\eta_L \geq \eta_{L,\text{min}}, \eta_{\text{esc}} \leq \eta_{\text{esc,max}}\}, \quad (15.10)$$

$$\mathcal{D}_{\text{imp}} = \{\Theta_I : \delta_{\text{imp}}(\Theta_I) < \epsilon_{\text{imp}}\}, \quad (15.11)$$

$$\mathcal{D}_J^{\text{num}} = \left\{ \Theta_I : \frac{|J_{\text{before}} - J_{\text{after}}|}{|J_{\text{before}}|} < \epsilon_J^{\text{num}} \right\}, \quad (15.12)$$

$$\mathcal{D}_J^{\text{EM}} = \left\{ \Theta_I : \frac{|J_{\oplus D,1} - J_{\text{loss}} - L_{\text{EM}}|}{L_{\text{EM}}} < \epsilon_J \right\}, \quad (15.13)$$

$$\mathcal{D}_{\text{accr}} = \{\Theta_I : M_{L,\text{calc}} \in [M_{L,\text{min}}, M_{L,\text{max}}]\}, \quad (15.14)$$

$$\mathcal{D}_{\text{geo}} = \left\{ \Theta_I : X_{\text{FeO}}^L, \phi_{\text{met},L} \text{ and siderophiles in the observed intervals} \right\}, \quad (15.15)$$

$$\mathcal{D}_{\text{iso}} = \left\{ \Theta_I : |R_X^L - R_X^{\oplus}| < \delta_X, X \in \{\text{O, Ti, Cr, Si, W}\} \right\}, \quad (15.16)$$

$$\mathcal{D}_{\text{chrono}} = \{\Theta_I : \text{order (5.2) respected and } t_L^{\text{accr}} \text{ compatible with the age of the Moon}\}, \quad (15.17)$$

and $\mathcal{D}_{\text{arch}}$ is given by Eq. (17.1).

Gate $\mathcal{D}_{\text{imp}}$. We write $\delta_{\text{imp}}(\Theta_I)$ for the relative deviation between the closed form (A.4) and direct integration on the hyperbolic conic, and ϵ_{imp} for the adopted threshold. This gate checks that the rectilinear approximation used for the closed form of the impulse remains acceptable within the viable domain. It is passed when the deviation between form (A.4) and direct integration on the hyperbolic conic remains below a fixed threshold, for example 10 % on the fraction η_L . This threshold is a modelling choice, to be justified according to the precision sought on the injection budget. If the gate is not passed, the corresponding branch of the viable domain is excluded, and the calculation must be repeated with the exact hyperbolic closed form (direct numerical integration).

Three clarifications on these gates. The first separates two requirements previously merged: $\mathcal{D}_J^{\text{num}}$ checks that the integrator conserves total angular momentum, which validates the calculation; $\mathcal{D}_J^{\text{EM}}$ checks that the state left by the passage reaches L_{EM} at the end of the evolution. For the second to be falsifiable, J_{loss} must be an *output* of Phase C, produced by the integrated removal mechanism, and not a parameter adjusted a posteriori.

The second concerns the integrity of B . The non-contact condition, $q_B > R_{\text{eq}} + R_{B,\text{eff}} \simeq 1.7 R_{\oplus}$ for $M_B = 0.1 M_{\oplus}$, is far less restrictive than tidal disruption, $d_{R,B} \simeq 2.75 R_{\oplus}$ from Eq. (6.21). They therefore deserve to remain separate, and the factor $\chi_{\text{tide}} > 1$ sets the safety margin: $\chi_{\text{tide}} = 1.1$ gives $q_B > 3.03 R_{\oplus}$ and $\chi_{\text{tide}} = 1.2$ gives $q_B > 3.30 R_{\oplus}$. The lower bound of Table 15.1 must be chosen consistently with χ_{tide} ; the most grazing columns of the pilot map are then set aside, which is the expected behaviour of a caution margin.

The third concerns $\mathcal{D}_{\text{accr}}$: the mass to be compared is $M_{L,\text{calc}}$ in the sense of Eq. (16.11), that is the sum of the sorting term and the injection term, and not the product $\eta_L M_{D,0}$ alone.

Falsification criterion

The theory requires a continuous region of parameters satisfying all gates together. A single, extremely tuned case does not constitute a robust domain of plausibility.

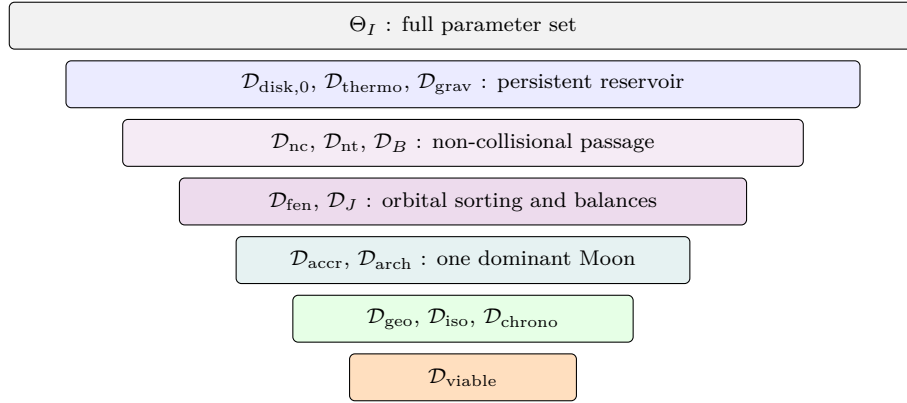


Figure 15.1: The viable domain as the intersection of successive gates. Only configurations that pass all gates define the physical path.

Chapter 16

Level I numerical program

The separation of time scales (Chapter 5) allows the calculation to be divided into three linked phases, throughout which each matter element preserves its dynamical, thermal and chemical labels.

16.1 Phase A: disk formation and evolution before B

The first simulation computes

$$\mathcal{A}(t) = \left\{ \dot{M}_{\text{in}}, \Sigma(r, t), T(r, t), P(r, t), L_D(t), R_{\text{out}}(t), \mathbf{X}(r, t), \mathbf{f}_{\text{phase}}(r, t), Q_T(r, t), \mathcal{G}(t) \right\}. \quad (16.1)$$

Its output is a family of states $\mathcal{D}_0$ accessible at time t_B .

16.2 Phase B: grazing passage of B

The encounter, integrated with the exact field (6.9) and closed in the barycentric frame, provides

$$\mathcal{B} = \{M_{D,1}, L_{D,1}, \Delta \mathbf{S}_{\oplus}, \Delta \mathbf{L}_{\text{orb}}, \Delta \mathbf{S}_B, M_{\oplus \rightarrow D}, M_{B \rightarrow D}, M_{\text{esc}}, i_{D,1}, e_D(r), \chi_J, \mathcal{H}_{B,1}\}. \quad (16.2)$$

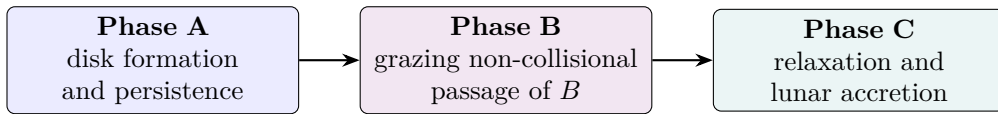
The distribution of particles or fluid elements is summarized by

$$\Pi_{\text{dyn}} = \Pi_{\text{dyn}}(\varepsilon_{\text{orb}}, j_z, q_p, i, r_0, \lambda_{\Omega}, z \mid \Theta_I). \quad (16.3)$$

16.3 Phase C: relaxation and accretion

The final phase computes

$$\mathcal{C} = \{M_L, L_L, i_{L,0}, \mathbf{X}_L, t_{\text{acc},L}, N_{\text{sat}}, f_{\text{dom}}, M_{\text{reaccr}}, M_{\text{esc}}\}. \quad (16.4)$$



The same matter elements preserve their dynamical, thermal and chemical labels throughout the chain.

Figure 16.1: Architecture of the numerical program of the theory.

16.4 Phase B protocol and viability maps

Model. Reservoir elements are first treated as test particles in the field of the proto-Earth, while B follows its exact hyperbola. The integration uses the differential acceleration (6.9) and stops when criterion (6.22) is satisfied. Each element is then classified according to $(\varepsilon, j_z, e, q_p, r_{\text{circ}})$ and criterion (8.1).

Cross-validation of the impulse. Phase B computes $\Delta \mathbf{v}$ by direct integration of the differential acceleration (6.9) on the true hyperbolic conic of B . The result is compared with the rectilinear closed form of Appendix A.6, and the deviation is quantified as a function of e_B and r/q_B . This deviation is an *output* of Phase B, not an adjusted parameter. It feeds gate $\mathcal{D}_{\text{imp}}$.

Response kernel. To decouple the response from the density profile, rings of elements are launched at about thirty initial radii r_0 between 2 and $5 R_\oplus$, with several hundred azimuths each. This gives the probability $P_k(r_0)$ that an element starting from r_0 ends in category $k \in \{L, \text{in}, \text{reaccr}, \text{esc}, \rightarrow B\}$. For any profile $\Sigma(r_0)$,

$$\eta_k = \frac{1}{M_{D,0}} \int_{R_{\text{in}}}^{R_{\text{out}}} \Sigma(r_0) P_k(r_0) 2\pi r_0 \, dr_0. \quad (16.5)$$

A single integration per set of B parameters can therefore be used for all reservoir profiles.

Reduced variables. The response depends on μ_B , $q_B/R_\oplus$, v_∞/v_{esc} , i_B and ω_B . The angular momentum ceded by the reservoir is obtained by weighting the elements; closure (9.5) assigns the opposite to the orbit of B , which remains valid as long as $M_D \ll M_B$.

Maps to produce. The viability maps are the main outputs of the program. They include the fractions $\Delta\eta_L(q_B, i_B)$, $\eta_{\text{esc}}(q_B, i_B)$ and $\eta_{\text{reaccr}}(q_B, i_B)$ for several values of μ_B and v_∞ ; the exchanged fraction χ_J ; the latitude-depth distribution of injected matter, $dM_{\oplus \rightarrow D}/d\lambda_\Omega \, dz$; and the inclination $i_{D,1}$ of the reorganized reservoir. These maps are the results to be computed; Figure 8.1 gives the analytical anticipation. The same phase must preserve the post-passage heliocentric orbit $\mathcal{H}_{B,1}$ so that a later calculation can test the gate $\mathcal{D}_{B,\text{hel}}$.

16.5 Injection budget: evaluating $M_{\oplus \rightarrow D}$

The injection channel introduced in Chapter 7 brings to the disk terrestrial surface material already depleted in metallic iron, nickel, cobalt and siderophile elements. It is the only term of balance (16.11) that is a *source*: it does not draw on $M_{D,0}$ and is therefore not bounded by η_L . It carries both the mass and the geochemical signature, which makes it the most rewarding quantity to evaluate. The following protocol computes it in four steps.

Step 1 — departure criterion. A surface element at colatitude θ leaves the proto-Earth when the sum of the centrifugal acceleration associated with $\Omega_\oplus$ and the radial component of the differential field (6.9) exceeds local gravity. Threshold (7.2) gives this condition at the sub- B point; it must be evaluated over the whole surface and at every instant, since the active zone moves during the passage.

Step 2 — angular momentum carried away. Each element leaves with

$$j_0 = \Omega_\oplus R_{\text{eq}}^2 \sin^2 \theta + \Delta j_B, \quad (16.6)$$

where Δj_B is the tangential contribution of B . The window criterion (8.1) then applies unchanged: $r_{\text{circ}} = j_0^2/GM_\oplus \geq a_{\text{Roche}}$. This step is decisive, since it decides whether the stripped matter reaches the accretion region or falls back. The ratio to be bridged is

$$\frac{j(a_{\text{Roche}})}{j(R_{\text{eq}})} = \sqrt{\frac{a_{\text{Roche}}}{R_{\text{eq}}}} \simeq 1.55 \quad (R_{\text{eq}} = 1.20 R_\oplus, \, a_{\text{Roche}} = 2.9 R_\oplus), \quad (16.7)$$

whereas the tidal impulse (8.11) supplies only about $0.04 v_K$ at $r = R_{\text{eq}}$ for $\mu_B = 0.1$ and $q_B = 3 R_\oplus$. The intrinsic rotation must therefore supply most of j_0 , with B crossing the last threshold: this is why $\tilde{\omega}$ is the dominant parameter of this budget, and why $M_{\oplus \rightarrow D}$ is highly sensitive to it.

Step 3 — integration. The surface is tiled in (θ, φ) and in depth, each element receiving its labels (λ_Ω, z) . Elements crossing the threshold are integrated in the combined field of the oblate proto-Earth and of B , along the hyperbolic trajectory, until the end of the encounter.

Step 4 — outputs. The calculation produces $M_{\oplus \rightarrow D}$ and its partition between the fraction beyond Roche, $M_{\oplus \rightarrow L}$, and the Roche-interior fraction, which feeds η_{in} ; the distribution in r_{circ} ; the angular momentum carried away, to be entered in balance (9.5); and the distribution in depth and latitude, which feeds directly the W/V ratio of Eq. (14.7).

Where the injected matter lands. A framing remark, which changes how the expected results are to be read. The circularization radius of an element leaving the surface in corotation is, to leading order,

$$r_{\text{circ}} \simeq \tilde{\omega}^2 R_{\text{eq}}, \quad (16.8)$$

since its specific angular momentum is $\Omega_\oplus R_{\text{eq}}^2$ at the equator. Even at the rotation limit, $\tilde{\omega} = 1$, the injected matter therefore circularizes near R_{eq} , that is 1.2 to 1.3 $R_\oplus$, and not beyond a_{Roche} . The tidal impulse supplied by B at $r = R_{\text{eq}}$, which remains of order a few hundredths of v_K according to Eq. (A.4), does not bridge the gap.

We propose not to read this result as a loss. The injected matter is neither ejected nor re-accreted: it joins the Roche-interior disk, that is the fraction η_{in} of partition (8.5), where it is available for the angular-momentum transport described in Section 10.3. The complete chain then becomes: B draws terrestrial surface material already depleted in metal, this material circularizes near 1.3 $R_\oplus$, it feeds the inner reservoir, and viscous or gravitational transport subsequently carries it beyond a_{Roche} where it accretes. The material lineage is the same from one end of the chain to the other, and the geochemical predictions of Chapter 14 are therefore unchanged: it is the same depleted matter that reaches the Moon, whatever route it takes to cross a_{Roche} .

The consequence is methodological. For the injection channel, η_L is not the right efficiency measure, since it counts *direct* placement in the lunar window. The relevant efficiency is that of spreading, that is the fraction of η_{in} that transport carries beyond a_{Roche} before the reservoir dissipates. Balance (16.11) must therefore be read with, for the term $M_{\oplus \rightarrow L}$, a spreading efficiency rather than a sorting efficiency.

Quantitative control

The sweep involves three parameters only, $\tilde{\omega}$, q_B and μ_B , the others being fixed by geometry. The expected sensitivity to $\tilde{\omega}$ is strong, and Eq. (16.8) indicates that the useful result of this calculation is not the fraction directly reaching a_{Roche} , but the total injected mass, its distribution in r_{circ} and the width of the rotation window that allows it, to be set against the corresponding spin in gate (15.13).

Falsification criterion

If, over the whole grid, the injected mass remains negligible compared with M_L , or if the spreading time needed to carry η_{in} beyond a_{Roche} exceeds the persistence duration (11.11), then the injection channel cannot compensate for a light reservoir, and only the massive branch of the reservoir remains open, with the associated angular-momentum constraints.

16.6 First pilot calculation of orbital sorting

A first Phase B calculation was performed to verify that the lunar window (8.1) can be reached over a non-punctual domain of parameters. It is not a hydrodynamical proof: the reservoir is represented by 1152 equal-mass test particles, initially circular and coplanar, distributed between 1.6 and 2.9 $R_\oplus$ with $\Sigma \propto r^{-1}$. The Roche limit is set to $a_{\text{Roche}} = 2.903 R_\oplus$, the equatorial radius

to $R_{\text{eq}} = 1.20 R_{\oplus}$, the mass of B to $M_B = 0.10 M_{\oplus}$, and the asymptotic velocity to $v_{\infty} = 2 \text{ km s}^{-1}$. The passage is prograde, with $\omega_B = 90^\circ$, and the acceleration used is the exact differential field of Eq. (6.9).

Table 16.1: Pilot map of the fraction $100\eta_L$ placed in the lunar window. Test-particle calculation, without disk self-gravity, without mutual collisions, without pressure and without hydrodynamics.

i_B	$q_B = 2.8 R_{\oplus}$	$3.0 R_{\oplus}$	$3.2 R_{\oplus}$	$3.4 R_{\oplus}$	$3.6 R_{\oplus}$
18°	13.5	11.4	9.7	9.0	7.3
21°	14.2	12.7	10.3	8.9	8.0
24°	15.7	12.7	11.2	9.4	7.8
27°	16.1	13.5	10.8	9.7	8.1
30°	17.1	14.5	11.3	10.3	8.7

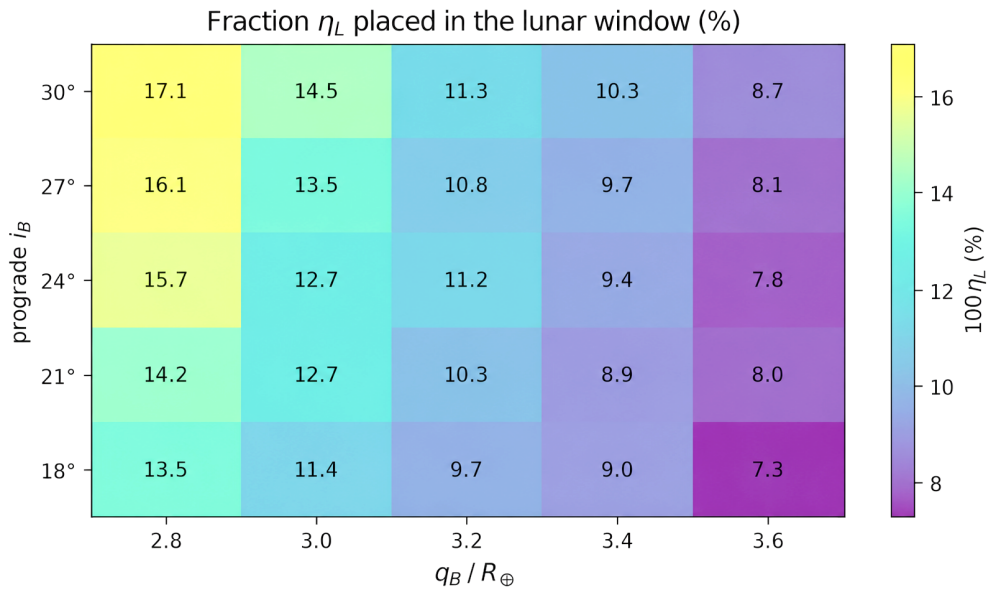


Figure 16.2: Pilot map of $\eta_L(q_B, i_B)$ for $M_B = 0.10 M_{\oplus}$, $v_{\infty} = 2 \text{ km s}^{-1}$ and $\omega_B = 90^\circ$. The map illustrates the existence of a continuous response band, to be confirmed by massive, collisional and self-gravitating simulations.

For the case $q_B = 3 R_{\oplus}$ and $i_B = 30^\circ$, the calculation gives

$$\eta_L = 0.145, \quad \eta_{\text{esc}} = 0.023, \quad \eta_{\text{reaccr}} = 0.019, \quad \eta_{\rightarrow B} = 0.023. \quad (16.9)$$

The remaining fraction, 0.789, remains bound in the inner domain or in states that do not immediately satisfy the lunar window. For a fixed initial mass, the mass directly placed in the window is

$$M_{L,\text{window}} \simeq 0.145 M_{D,0}, \quad (16.10)$$

which means that one lunar mass immediately placed in this window would require $M_{D,0} \simeq 6.9 M_L$ in this test calculation. For $q_B = 2.8 R_{\oplus}$ and $i_B = 30^\circ$, the value $\eta_L = 0.171$ lowers this mass to $M_{D,0} \simeq 5.85 M_L$.

These figures are sorting diagnostics, not a lunar formation budget. The quantity η_L measures an orbital sorting efficiency; the sorted matter must still survive relaxation and then accrete. The relevant mass balance is therefore

$$M_{L,\text{calc}} = \epsilon_{\text{acc}} \epsilon_{\text{surv}} \eta_L M_{D,0} + M_{\oplus \rightarrow L} + M_{B \rightarrow L} \pm \delta M_{L,\text{imp}}, \quad (16.11)$$

where $\delta M_{L,\text{imp}}$ is the systematic uncertainty introduced by the rectilinear approximation, estimated by comparing the closed form (A.4) with direct integration on the hyperbolic conic (gate (15.11)).

Two lessons follow. First, since $\epsilon_{\text{acc}}\epsilon_{\text{surv}} \leq 1$, any refinement of these efficiencies increases the reservoir mass required at fixed lunar mass, and hence also $L_{D,0}$, to which it is proportional: the gain is paid on gate $\mathcal{D}_J^{\text{EM}}$ (15.13). Second, and more importantly, the term $M_{\oplus \rightarrow L}$ does not depend on $M_{D,0}$ at all. It is through this term that the theory can satisfy $\mathcal{D}_{\text{accr}}$ with a modest reservoir, and it is therefore the quantity to evaluate first (Section 16.5).

Associated angular momentum. A reservoir of $6.9 M_L$ so structured carries, in Keplerian rotation,

$$L_{D,0} = 2\pi \int_{R_{\text{in}}}^{R_{\text{out}}} \Sigma(r) \sqrt{GM_{\oplus}} r \, dr \simeq 3.8 \times 10^{34} \text{ kg m}^2 \text{ s}^{-1} \simeq 1.09 L_{\text{EM}}, \quad (16.12)$$

to which the proto-Earth spin adds about $1.3 L_{\text{EM}}$ for a 3 h period. Two useful consequences. First, since $L_{D,0}$ is proportional to $M_{D,0}$, the light branch of the reservoir lightens the budget accordingly: one more reason to prefer it. Second, the passage does not have to supply the angular momentum of the final system, which is already available; its function is indeed to redistribute it. The state left by the passage must nevertheless reach L_{EM} at the end of tidal evolution, which requires an explicit removal mechanism in Phase C, for instance capture into solar evection resonance or a Laplace-plane transition [1, 2], both belonging to the classical celestial mechanics of the Earth–satellite system. This is the purpose of gate $\mathcal{D}_J^{\text{EM}}$ (15.13) and of prediction P8.

The retrograde control isolates the role of passage direction. For $q_B = 3 R_{\oplus}$, at identical periastron latitude and with the other parameters unchanged, one obtains

$$\eta_{L,\text{retro}} \simeq 0.027, \quad (16.13)$$

that is, an efficiency about 5.4 times lower than in the prograde case. This result is consistent with the coherence-duration argument of Eq. (6.16).

The radial origin of the matter promoted into the lunar window confirms that sorting acts mainly on the external part of the reservoir, close to Roche: for the case $q_B = 3 R_{\oplus}$, $i_B = 30^\circ$, the contributions to the η_L cohort come 35.0% from the $2.53\text{--}2.71 R_{\oplus}$ annulus and 53.1% from the $2.71\text{--}2.90 R_{\oplus}$ annulus.

Quantitative control

The pilot calculation provides a concrete positive signal: the lunar window is not empty in the tested region, and the prograde low-latitude configuration is much more efficient than the retrograde control. It does not yet prove viability, because self-gravity, collisions, hydrodynamics, thermal evolution and balance closure still have to be added. Two clarifications help to situate it. First its validity domain in mass: for $M_{D,0} = 6.9 M_L$, that is $0.085 M_{\oplus}$, the reservoir reaches $0.85 M_B$ and its self-gravity becomes of the same order as the measured forcing; the test-particle approximation is by contrast legitimate in the light branch, $M_{D,0} \simeq 2\text{--}3 M_L$, toward which control (12.8) and criterion (11.9) both point. Table 16.1 is therefore to be read as a trend map in (q_B, i_B) , at fixed mass. Second its validity domain in mechanism: the 1152 particles represent the preexisting reservoir alone, with no matter injected from the terrestrial surface; the map measures η_L , that is the first term of Eq. (16.11), and leaves $M_{\oplus \rightarrow D}$ entirely to be evaluated. Its role is to identify the promising region of parameter space to be explored by Level I.

Falsification criterion

If, on the full grid, $\Delta\eta_L$ remains negligible, or only increases at the cost of an escaped fraction incompatible with the reservoir mass, the orbital-sorting mechanism is insufficient and the corresponding branch is excluded.

Didactic reminder

The viability domain fixes what the calculation must demonstrate. The final part states what nature should show if the theory is correct, and how to verify it.

Part IX

Predictions

Each prediction is stated in the same form: what should be observed *if the theory is correct*, and then *how to verify it*, through measurement of lunar samples or through calculation. The predictions concern architecture, composition, chronology and dynamics.

Chapter 17

Predictions and verification

Prediction — P1. One dominant satellite

If the theory is correct. The sorted and relaxed reservoir produces a satellite mass almost entirely concentrated in one body:

$$\mathcal{D}_{\text{arch}} = \{\Theta_I : M_L \simeq M_{L,\text{obs}}, N_{\text{sat},\text{large}} \simeq 1, f_{\text{dom}} \rightarrow 1\}. \quad (17.1)$$

How to verify it. Phase C N -body simulations, with collisions and self-gravity, starting from the reservoirs reorganized in Phase B; tracking of N_{sat} and f_{dom} .

Prediction — P2. A negligible share of B

If the theory is correct. The fraction of lunar matter derived from B ,

$$f_B = \frac{M_{B \rightarrow L}}{M_L}, \quad (17.2)$$

is small, and the oxygen, titanium and chromium isotopic compositions of the Moon are indistinguishable from those of the silicate Earth within the limits (14.5).

How to verify it. High-precision isotopic measurements on varied lunar samples; calculation of $M_{B \rightarrow D}$ in Phase B for passages with q_B close to $d_{R,B}$.

Prediction — P3. Terrestrial tungsten

If the theory is correct. After correction for late accretion, lunar $\varepsilon^{182}\text{W}$ is identical to that of Earth's mantle, and the Hf/W ratio of lunar matter does not produce a ^{182}W excess incompatible with time t_B .

How to verify it. Measurements of $\varepsilon^{182}\text{W}$ and Hf/W on lunar rocks with low cosmic-ray exposure; comparison with the Phase A chronology.

Prediction — P4. Inheritance from terrestrial core segregation

If the theory is correct. Lunar depletions in V, Cr and Mn reproduce those of Earth's mantle, and the lunar W/U ratio is close to the terrestrial ratio.

How to verify it. Estimates of lunar-mantle composition from basalts and primitive rocks; high-pressure metal-silicate partitioning experiments.

Prediction — P5. A dynamical-chemical correlation

If the theory is correct. The same matter cohort carries both an orbital history and a composition:

$$(r_0, \lambda_\Omega, z) \longleftrightarrow (\varepsilon_{\text{orb}}, j_z, q_p, \mathbf{X}), \quad (17.3)$$

and the W/V ratio of injected matter (14.7) reflects the depth and latitude from which it was stripped.

How to verify it. Tracking of the labels (λ_Ω, z) in Phase B; comparison of computed W/V with measured ratios normalized to elements of similar behavior.

Prediction — P6. A reservoir of terrestrial composition

If the theory is correct. The preexisting circumterrestrial reservoir is terrestrial silicate in nature, fed mainly by equatorial mass loss and mantle ejecta, and the lunar FeO excess is produced by internal transformations within the reservoir.

How to verify it. Phase A balances on the channels (3.8); calculation of Π_D and comparison with lunar FeO and siderophile contents.

Prediction — P7. A non-zero initial inclination

If the theory is correct. A slightly inclined passage gives the reorganized reservoir an angular momentum not collinear with Earth's spin:

$$i_{D,1} = \arccos \left(\frac{\mathbf{L}_{D,1} \cdot \mathbf{S}_{\oplus,1}}{|\mathbf{L}_{D,1}| |\mathbf{S}_{\oplus,1}|} \right). \quad (17.4)$$

This quantity constitutes an initial condition for the tidal evolution of the Earth–Moon system, whose lunar orbital inclination is one of the classical enigmas [32].

How to verify it. Calculation of $i_{D,1}$ in Phase B, then integration of tidal evolution up to the present orbit.

Prediction — P8. Compatible angular momentum after removal

If the theory is correct. The angular momentum of the Earth–reservoir system left by the passage, $J_{\oplus D,1}$, exceeds L_{EM} the more so as the reservoir is massive and the proto-Earth fast. The theory therefore predicts that a post-accretional removal is necessary, and that the pair $(J_{\oplus D,1}, i_{D,1})$ produced by the passage lies in the capture domain of a known mechanism.

How to verify it. Closure of balance (9.5) in Phase B, then tidal integration of $J_{\oplus D,1}$ down to L_{EM} , with J_{loss} an output of the calculation by evection resonance or Laplace-plane transition [1, 2]. If no pair $(J_{\oplus D,1}, i_{D,1})$ of the viable domain can be captured, gate (15.13) closes on the corresponding branch.

Prediction — P9. Efficient orbital sorting in the dynamical low-latitude band

If the theory is correct. There exists a continuous region of the prograde low-latitude band in which $\Delta\eta_L$ is significant and η_{esc} moderate, in accordance with (8.8), while passing gates $\mathcal{D}_f^{\text{EM}}$, $\mathcal{D}_{\text{grav}}$, $\mathcal{D}_{\text{disk},0}$, $\mathcal{D}_{\text{accr}}$, $\mathcal{D}_{\text{tide},B}$ and $\mathcal{D}_{\text{imp}}$.

How to verify it. Phase B viability maps (Chapter 16), crossed with the injection budget of Section 16.5 and the angular-momentum balance.

Prediction — P10. An eliminated heliocentric fate for B

If the theory is correct. Planet B belongs to the population of primordial planetary bodies later eliminated:

$$B \in \mathcal{P}_{\text{elim}}, \quad t_B < t_{\text{ej}}, \quad (17.5)$$

and the post-passage heliocentric orbit $\mathcal{H}_{B,1}$ is compatible with later diffusion toward collision, accretion by a planet or ejection from the Solar System.

How to verify it. Heliocentric N -body integrations starting from $\mathcal{H}_{B,1}$, with both early and late instability scenarios, and verification that the final architectures remain compatible with the present Solar System.

General conclusion

The question asked at the beginning of this work was that of a path: how could the Earth–Moon system observed today have emerged from the proto-terrestrial system through a physically coherent evolution?

The theory proposes a precise answer. During its growth, the proto-Earth is accompanied by a secondary circumterrestrial reservoir, analogous in principle to the accretion systems of giant planets and to the disk of PDS 70c, but dominated by solids and fed mainly by its own silicate layer. This reservoir already contains mass and angular momentum when planet B passes.

B does not supply the bulk of the lunar matter. It is a transient gravitational agent redistributing mass and angular-momentum fluxes between the reservoirs. Grazing passages are at least as frequent as collisions (Eq. (6.2)); the trajectory of B carries an angular momentum of the same order as that of the Earth–Moon system (Eq. (6.18)); in the low-latitude band of an oblate, rapidly rotating proto-Earth, its prograde passage injects superficial matter already depleted in metal. Its central result is orbital sorting: a bound cohort is placed by the torque of B in the lunar window beyond the Roche limit (Eqs. (8.1) and (8.3)), while other fractions re-accrete or escape.

The sorted reservoir relaxes and accretes the Moon, concentrating satellite mass into a dominant body. Geochemistry belongs to the same story: the non-collisional passage favors a small contribution from B , making isotopic similarity an expected constraint of the viable domain; vanadium and tungsten, read together, trace the fluxes of terrestrial silicate matter and the depth of the sampled layer.

The logic of the model closes with the fate of B . The agent that reorganizes the circumterrestrial reservoir need not remain in the final Solar System: it may belong to the population of primordial planetary bodies later eliminated by the global orbital instability, provided that its passage satisfies $t_B < t_{ej}$. This hypothesis links Earth–Moon formation to migration and instability in the primordial Solar System without requiring B to be identified with a single specific missing planet.

The final scientific question is

$$\boxed{\mathcal{S}_0 = \{\Theta_I\} \longrightarrow \mathcal{D}_0 \xrightarrow{\text{orbital sorting by } B} \mathcal{D}_1 \longrightarrow (M_L, L_L, i_{L,0}, \mathbf{X}_L) \in \mathcal{D}_{\text{viable}} \longrightarrow \mathcal{S}_f.} \quad (17.6)$$

The theory does not claim to demonstrate a unique history. It defines a quantitative and falsifiable physical path, and indicates, for each of its stages, how to confirm, restrict or exclude it. The transport law by grazing passage is an *analytical form under hypotheses*, whose validation on the hyperbolic conic is an output of Phase B and not an adjusted parameter. It is this validation that transforms the static threshold into a closed model.

Appendix A

Demonstrations

A.1 Circularization radius of captured matter

Matter captured through an accretion zone of radius R_{acc} in a sheared flow of frequency Ω carries a specific angular momentum of order $j \simeq \Omega R_{\text{acc}}^2/4$ [24]. It circularizes at the radius where its angular momentum equals the Keplerian value, $j^2 = GM_{\oplus} r_c$:

$$r_c = \frac{\Omega^2 R_{\text{acc}}^4}{16 GM_{\oplus}}. \quad (\text{A.1})$$

For $R_{\text{acc}} = R_H$, the definition of the Hill radius, $R_H^3 = GM_{\oplus}/(3\Omega^2)$, gives $\Omega^2 = GM_{\oplus}/(3R_H^3)$ and $r_c = R_H/48$. For accretion limited to the Bondi sphere, r_c is multiplied by $(R_B/R_H)^4$, leading to Eq. (3.7).

A.2 Frequency of grazing passages

For a hyperbolic trajectory with impact parameter b , conservation of angular momentum and energy gives $b v_{\infty} = q v_p$ and $v_p^2 = v_{\infty}^2 + 2GM_{\text{tot}}/q$, hence

$$b^2 = q^2 \left(1 + \frac{2GM_{\text{tot}}}{q v_{\infty}^2} \right), \quad \sigma(q) = \pi b^2, \quad (\text{A.2})$$

which is Eq. (6.1). With $\Theta = 2GM_{\text{tot}}/(R_c v_{\infty}^2)$, $\sigma(R_c) = \pi R_c^2(1 + \Theta)$ and $\sigma(2R_c) = \pi R_c^2(4 + 2\Theta)$; the ratio $[\sigma(2R_c) - \sigma(R_c)]/\sigma(R_c) = (3 + \Theta)/(1 + \Theta)$ decreases from 3 to 1 as Θ increases from 0 to infinity.

A.3 Trajectory nodes and non-crossing

The equation of the conic is $r = p/(1 + e \cos \nu)$ with $p = q(1 + e)$. The trajectory crosses the reference plane when the argument of latitude $\omega + \nu$ is 0 or π , i.e. $\nu = -\omega$ or $\nu = \pi - \omega$; the corresponding radii are $p/(1 + e \cos \omega)$ and $p/(1 - e \cos \omega)$, whose smaller value is $p/(1 + e|\cos \omega|)$. This is Eq. (6.6).

A.4 Circularization radius and perigee

For a bound orbit, $j^2 = GM_{\oplus} a(1 - e^2)$ and $q_p = a(1 - e)$, hence $r_{\text{circ}} = j^2/(GM_{\oplus}) = a(1 - e)(1 + e) = q_p(1 + e)$. Since $0 \leq e < 1$, $q_p \leq r_{\text{circ}} < 2q_p$, which establishes Eq. (8.2).

A.5 Orbital-sorting boundaries

An element on a circular orbit of radius r_0 has velocity $v_K = \sqrt{GM_{\oplus}/r_0}$. After a tangential impulse, $v = v_K(1 + \delta)$ and $j = r_0 v_K(1 + \delta)$, hence $r_{\text{circ}} = r_0(1 + \delta)^2$. Its specific energy is

$$\varepsilon = \frac{GM_{\oplus}}{r_0} \left[\frac{(1 + \delta)^2}{2} - 1 \right], \quad (\text{A.3})$$

which vanishes for $(1 + \delta)^2 = 2$: $\delta_{\text{esc}} = \sqrt{2} - 1$. For $\delta < 0$, r_0 becomes the apogee; the semi-major axis is $a = r_0/[2 - (1 + \delta)^2]$ and the perigee is $q_p = 2a - r_0 = r_0(1 + \delta)^2/[2 - (1 + \delta)^2]$. The condition $q_p = R_{\text{eq}}$ gives $(1 + \delta)^2 = 2R_{\text{eq}}/(r_0 + R_{\text{eq}})$, hence δ_{re} . The condition $r_{\text{circ}} = a_{\text{Roche}}$ gives $\delta_L = \sqrt{a_{\text{Roche}}/r_0} - 1$. This gives Eq. (8.10).

A.6 Tidal impulse: analytical form under hypotheses

Status of this result. This section establishes an *analytical reference form* for the tidal impulse received by a reservoir element during the passage of B . It is not exact in the strict sense: it rests on three explicit hypotheses, stated below. Its domain of validity is specified, and Phase B of the numerical program is responsible for validating it by direct integration on the true hyperbolic conic.

Hypotheses.

1. **Rectilinear trajectory.** The trajectory of B is assimilated to a straight line of velocity v_p and impact parameter q_B throughout the passage.
2. **Impulsive forcing.** The encounter duration is short compared with the local orbital period: $\eta(r) = \Omega_K(r) t_{\text{enc}} \ll 1$.
3. **Fixed particle.** The matter element does not move during the passage; its own orbital motion is neglected compared with the received impulse.

Conventions. Geocentric Cartesian frame $(O; x, y, z)$: the x -axis is along the velocity of B at periastron, the y -axis is directed from the proto-Earth center toward B at periastron, the z -axis completes the right-handed triad. The nominal trajectory is $\mathbf{r}_B(t) = (v_p t, q_B, 0)$.

Derivation. The exact differential field is given by Eq. (6.9). Integrating the direct and indirect terms separately over the rectilinear trajectory gives the following closed form, valid for all $r < q_B$:

$$\Delta \mathbf{v}(\mathbf{r}) = -\frac{GM_B}{v_p q_B} \left(0, \frac{2(\eta - 1)}{(\eta - 1)^2 + \zeta^2} + 2, \frac{2\zeta}{(\eta - 1)^2 + \zeta^2} \right) \quad (\text{A.4})$$

with the reduced coordinates $\eta = y/q_B$ and $\zeta = z/q_B$.

The first-order approximation in r/q_B follows:

$$\Delta \mathbf{v}^{\text{approx}}(\mathbf{r}) = -\frac{2GM_B}{v_p q_B^2} (0, y, -z) + O\left(\frac{r^2}{q_B^3}\right). \quad (\text{A.5})$$

This is the expression historically known in the literature as the impulsive approximation.

Comparison and domain of validity. Table A.1 compares the two forms for a particle located on the impact parameter axis ($\zeta = 0$) and on the perpendicular axis ($\eta = 0$). The relative deviation is computed with respect to the closed form (A.4).

Table A.1: Deviation between the closed form (A.4) and the first-order approximation (A.5).

r/q_B	Deviation on the impact parameter axis	Deviation on the perpendicular axis
0.1	+11 %	−1 %
0.3	+43 %	−8 %
0.5	+100 %	−20 %
0.7	+233 %	−33 %
0.9	+900 %	−45 %

The first-order approximation loses all precision once $r/q_B \gtrsim 0.3$. For the reference case of the manuscript ($q_B = 3 R_\oplus$, reservoir up to $\sim 2.9 R_\oplus$), the ratio r/q_B varies from 0.53 to 0.97: the closed form (A.4) must therefore be used as the reference, and the first-order approximation reserved for order-of-magnitude checks.

Order-of-magnitude bound. In the limit $r \ll q_B$, the closed form recovers the estimate used in the main text:

$$\frac{\Delta v}{v_K} \sim \sqrt{2} \frac{\mu_B}{\sqrt{1 + \mu_B}} \left(\frac{r}{q_B} \right)^{3/2}. \quad (\text{A.6})$$

This bound is useful for fixing orders of magnitude; it does not replace form (A.4).

Residual limitation. The closed form (A.4) still assumes a rectilinear trajectory. For the reference case ($e_B \simeq 1.1$), curvature is not negligible and direct integration on the true hyperbolic conic is required. This integration is performed in Phase B, and the deviation from the closed form is an *output* of the calculation, quantified as a function of e_B and r/q_B . It is this cross-validation that authorizes presenting the transport law as *analytical under hypotheses*, and not as exact in the strict sense.

A.7 Circularization energy

A parcel with energy $\varepsilon_i = -GM_\oplus/(2a)$ circularizes at conserved angular momentum at radius $a_{\text{circ}} = a(1 - e^2)$, with energy $\varepsilon_f = -GM_\oplus/(2a_{\text{circ}})$. The released energy is $\varepsilon_i - \varepsilon_f = \frac{GM_\oplus}{2a} \left[\frac{1}{1-e^2} - 1 \right] = \frac{GM_\oplus}{2a} \frac{e^2}{1-e^2}$, which is Eq. (12.4).

A.8 Injection threshold

At the sub- B point, the radial tidal acceleration is, for $\lambda_B = 0$,

$$GM_B \left[(q_B - R_{\text{eq}})^{-2} - q_B^{-2} \right] \simeq 2GM_B R_{\text{eq}}/q_B^3. \quad (\text{A.7})$$

At low latitude, its projection on the radial direction is multiplied by $F(\lambda_B) = (3 \cos^2 \lambda_B - 1)/2$. Equating it to the effective gravity $(GM_\oplus/R_{\text{eq}}^2)(1 - \tilde{\omega}^2)$ gives

$$1 - \tilde{\omega}^2 \lesssim 2F(\lambda_B) \frac{M_B}{M_\oplus} \left(\frac{R_{\text{eq}}}{q_B} \right)^3, \quad (\text{A.8})$$

which is Eq. (7.2).

A.9 Bound on the chondritic share of the reservoir

With $f_B = 0$ and $\varepsilon^{182}\text{W}_\oplus = 0$, Eq. (13.3) becomes $\varepsilon^{182}\text{W}_L = f_0 C_{W,0} \varepsilon^{182}\text{W}_0 / [f_0 C_{W,0} + (1 - f_0) C_{W,\oplus}]$. For $f_0 C_{W,0} \ll C_{W,\oplus}$, the denominator reduces to $C_{W,\oplus}$, and the condition $|\varepsilon^{182}\text{W}_L| < \delta \varepsilon$ gives Eq. (14.1).

Appendix B

Notation

Symbol	Meaning
$\mathcal{S}_0, \mathcal{S}_f$	initial state of the proto-terrestrial system, observed state
$\mathcal{R}_\oplus, \mathcal{R}_D, \mathcal{R}_B, \mathcal{R}_\infty$	reservoirs: proto-Earth, circumterrestrial reservoir, planet B , external medium
$\Phi_{i \rightarrow j}$	mass flux from reservoir i to reservoir j
$\mathcal{T}_{j \rightarrow i}$	torque exerted by reservoir j on reservoir i
$\mathbf{X}_i, \mathbf{X}_{i \rightarrow j}$	composition of a reservoir, composition of a flux
$M_\oplus, R_{\text{eq}}, \Omega_\oplus, S_\oplus$	mass, equatorial radius, angular velocity and spin of the proto-Earth
$\tilde{\omega}$	reduced rotation $\Omega_\oplus / \sqrt{GM_\oplus / R_{\text{eq}}^3}$
J_2	gravitational flattening coefficient
$M_D, L_D, \Sigma(r)$	mass, angular momentum and surface density of the reservoir
$\mathcal{D}_0, \mathcal{D}_1$	reservoir state before and after the encounter
$\mathcal{G}(t)$	population of aggregates and moonlets
$R_{\text{in}}, R_{\text{out}}$	inner and outer radii of the reservoir
R_H, R_B	Hill and Bondi radii
$a_{\text{Roche}}, a_{\text{sync}}$	Roche limit, synchronous orbit
M_B, μ_B, R_B, ρ_B	mass, relative mass, radius and density of B
q_B, v_∞, v_p, e_B	periastron, asymptotic velocity, periastron velocity, eccentricity of B 's trajectory
i_B, ω_B	inclination relative to the rotational equator, argument of periastron
λ_B	latitude of the sub- B point at periastron
λ_c	half-width of the dynamical low-latitude band
$t_{\text{enc}}, \eta(r)$	encounter duration, regime parameter $\Omega_K t_{\text{enc}}$
$d_{R,B}$	Roche distance of B around the proto-Earth
$L_{B,\text{rel}}, \chi_J$	relative orbital angular momentum of B , exchanged fraction
$\varepsilon, j, j_z, e, q_p$	energy, angular momentum, projection, eccentricity and perigee of an element
r_{circ}	circularization radius $j^2 / (GM_\oplus)$
$\eta_L, \Delta\eta_L$	fraction in the lunar window, specific contribution of B
$\eta_{\text{in}}, \eta_{\text{reaccret}}, \eta_{\text{esc}}, \eta_{\rightarrow B}$	fractions: inner disk, re-accretion, escape, capture by B
δ	relative tangential impulse $\Delta v_t / v_K$
$f_0, f_\oplus, f_B$	provenance fractions of lunar matter
$f_{\text{dom}}, N_{\text{sat}}$	dominance factor, number of satellites
λ_Ω, z	dynamical latitude and depth of origin of injected matter
$\mathcal{P}_{\text{elim}}$	population of primordial planetary bodies later eliminated by collision, accretion, diffusion or ejection
t_{ej}	time of ejection or dynamical elimination of B
$\mathcal{H}_{B,1}$	heliocentric orbit of B after its passage near the proto-Earth
Θ_I	Level I parameter vector
$\mathcal{D}_{\text{viable}}$	viability domain
$\mathcal{D}_{B,\text{hel}}$	heliocentric gate checking the eliminated fate of B
$\mathcal{D}_{\text{imp}}$	gate validating the impulse on the hyperbolic conic

Appendix C

Constants and numerical values

Table C.1: Constants and reference values used in the work.

Quantity	Symbol	Value
Terrestrial gravitational parameter	$GM_{\oplus}$	$3.986 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$
Earth mass	$M_{\oplus}$	$5.972 \times 10^{24} \text{ kg}$
Mean Earth radius	$R_{\oplus}$	$6.371 \times 10^6 \text{ m}$
Moon mass	M_L	$7.342 \times 10^{22} \text{ kg}$
Solar mass	$M_{\odot}$	$1.989 \times 10^{30} \text{ kg}$
Astronomical unit	$a_{\oplus}$	$1.496 \times 10^{11} \text{ m}$
Mean lunar density	ρ_s	3.34 g cm^{-3}
Mean terrestrial density	$\bar{\rho}_{\oplus}$	5.51 g cm^{-3}
Earth–Moon angular momentum	L_{EM}	$\approx 3.5 \times 10^{34} \text{ kg m}^2 \text{ s}^{-1}$
Roche limit ($\rho_s = 3.34$)	a_{Roche}	$\approx 2.9 R_{\oplus}$
Half-life of ^{182}Hf	$t_{1/2}$	8.9 Ma

Bibliography

- [1] C. D. Murray and S. F. Dermott, *Solar System Dynamics*, Cambridge University Press, 1999.
- [2] S. Tremaine, J. Touma and F. Namouni, “Satellite dynamics on the Laplace surface”, *The Astronomical Journal*, 137 (2009), 3706–3717. doi:10.1088/0004-6256/137/3/3706.
- [3] S. A. Balbus and J. F. Hawley, “A powerful local shear instability in weakly magnetized disks. I. Linear analysis”, *The Astrophysical Journal*, 376 (1991), 214–222. doi:10.1086/170270.
- [4] N. I. Shakura and R. A. Sunyaev, “Black holes in binary systems. Observational appearance”, *Astronomy and Astrophysics*, 24 (1973), 337–355.
- [5] J. Frank, A. King and D. Raine, *Accretion Power in Astrophysics*, 3rd edition, Cambridge University Press, 2002.
- [6] J. E. Pringle, “Accretion discs in astrophysics”, *Annual Review of Astronomy and Astrophysics*, 19 (1981), 137–160. doi:10.1146/annurev.aa.19.090181.001033.
- [7] A. Johansen and P. Lacerda, “Prograde rotation of protoplanets by accretion of pebbles in a gaseous environment”, *Monthly Notices of the Royal Astronomical Society*, 404 (2010), 475–485. doi:10.1111/j.1365-2966.2010.16309.x.
- [8] R. G. Visser, C. W. Ormel, C. Dominik and S. Ida, “Spinning up planetary bodies by pebble accretion”, *Icarus*, 335 (2020), 113380. doi:10.1016/j.icarus.2019.07.014.
- [9] A. Johansen, T. Ronnet, M. Bizzarro, M. Schiller, M. Lambrechts, Å. Nordlund and H. Lammer, “A pebble accretion model for the formation of the terrestrial planets in the Solar System”, *Science Advances*, 7 (2021), eabc0444. doi:10.1126/sciadv.abc0444.
- [10] R. M. Canup and W. R. Ward, “A common mass scaling for satellite systems of gaseous planets”, *Nature*, 441 (2006), 834–839. doi:10.1038/nature04860.
- [11] A. Toomre, “On the gravitational stability of a disk of stars”, *The Astrophysical Journal*, 139 (1964), 1217–1238. doi:10.1086/147861.
- [12] S. Ida, R. M. Canup and G. R. Stewart, “Lunar accretion from an impact-generated disk”, *Nature*, 389 (1997), 353–357. doi:10.1038/38669.
- [13] J. Salmon and R. M. Canup, “Lunar accretion from a Roche-interior fluid disk”, *The Astrophysical Journal*, 760 (2012), 83. doi:10.1088/0004-637X/760/1/83.
- [14] R. M. Canup and W. R. Ward, “Formation of the Galilean satellites: conditions of accretion”, *The Astronomical Journal*, 124 (2002), 3404–3423. doi:10.1086/344684.
- [15] F. Nimmo, R. Canup, Y. Fujii, A. Izidoro, J. Ji, W. McKinnon, O. Mousis, C. Ormel, M. Ogihara, D. Stevenson and M. Blanc, “Origin and evolution of the Galilean satellites within the Jovian system”, *Space Science Reviews*, 222 (2026), article 46. doi:10.1007/s11214-026-01295-6.
- [16] S. Charnoz, J. Salmon and A. Crida, “The recent formation of Saturn’s moonlets from viscous spreading of the main rings”, *Nature*, 465 (2010), 752–754. doi:10.1038/nature09096.
- [17] M. Blanc, A. Crida, Y. Shibaike, S. Charnoz, M. El Moutamid, P. Estrada, O. Mousis, J. Salmon, A. Schneeberger and P. Vernazza, “Understanding the formation of Saturn’s regular moons in the context of giant planet moons formation scenarios”, *Space Science Reviews*, 221 (2025), article 35. doi:10.1007/s11214-025-01156-8.
- [18] R. Helled, N. Nettelmann and T. Guillot, “Uranus and Neptune: origin, evolution and internal structure”, *Space Science Reviews*, 216 (2020). doi:10.1007/s11214-020-00660-3.

- [19] M. Benisty, J. Bae, S. Facchini, M. Keppler, R. Teague, A. Isella, N. T. Kurtovic, L. M. Pérez, A. Sierra, S. M. Andrews et al., “A circumplanetary disk around PDS70c”, *The Astrophysical Journal Letters*, 916 (2021), L2. doi:10.3847/2041-8213/ac0f83.
- [20] Jet Propulsion Laboratory, Solar System Dynamics, “Planetary satellite physical parameters and mean elements”, JPL/Caltech, data accessed September 2026.
- [21] A. Popovas, Å. Nordlund, J. P. Ramsey and C. W. Ormel, “Pebble dynamics and accretion on to rocky planets – I. Adiabatic and convective models”, *Monthly Notices of the Royal Astronomical Society*, 479 (2018), 5136–5156. doi:10.1093/mnras/sty1752.
- [22] E. L. Ruskol, “The origin of the Moon. I. Formation of a swarm of bodies around the Earth”, *Soviet Astronomy*, 4 (1960), 657.
- [23] S. J. Weidenschilling, R. Greenberg, C. R. Chapman, F. Herbert, D. R. Davis, M. J. Drake, J. Jones and W. K. Hartmann, “Origin of the Moon from a circumterrestrial disk”, in W. K. Hartmann, R. J. Phillips and G. J. Taylor (eds.), *Origin of the Moon*, Lunar and Planetary Institute, Houston (1986), 519–550.
- [24] A. C. Quillen and D. E. Trilling, “Do proto-Jovian planets drive outflows?”, *The Astrophysical Journal*, 508 (1998), 707–713.
- [25] R. G. Martin and S. H. Lubow, “Tidal truncation of circumplanetary discs”, *Monthly Notices of the Royal Astronomical Society*, 413 (2011), 1447–1461.
- [26] C. W. Ormel, J.-M. Shi and R. Kuiper, “Hydrodynamics of embedded planets’ first atmospheres – II. A rapid recycling of atmospheric gas”, *Monthly Notices of the Royal Astronomical Society*, 447 (2015), 3512–3525.
- [27] J. Fung, P. Artymowicz and Y. Wu, “The 3D flow field around an embedded planet”, *The Astrophysical Journal*, 811 (2015), 101.
- [28] H. Daisaka, H. Tanaka and S. Ida, “Viscosity in a dense planetary ring with self-gravitating particles”, *Icarus*, 154 (2001), 296–312.
- [29] V. S. Safronov, “On the gravitational instability in flattened systems with axial symmetry and non-uniform rotation”, *Annales d’Astrophysique*, 23 (1960), 979.
- [30] P. Goldreich and D. Lynden-Bell, “Gravitational stability of uniformly rotating disks”, *Monthly Notices of the Royal Astronomical Society*, 130 (1965), 97–124.
- [31] E. Kokubo, S. Ida and J. Makino, “Evolution of a circumterrestrial disk and formation of a single moon”, *Icarus*, 148 (2000), 419–436.
- [32] K. Pahlevan and A. Morbidelli, “Collisionless encounters and the origin of the lunar inclination”, *Nature*, 527 (2015), 492–494. doi:10.1038/nature16137.
- [33] A. Breslau, M. Steinhausen, K. Vincke and S. Pfalzner, “Sizes of protoplanetary discs after star-disc encounters”, *Astronomy & Astrophysics*, 565 (2014), A130.
- [34] T. S. Kruijer, T. Kleine, M. Fischer-Gödde and P. Sprung, “Lunar tungsten isotopic evidence for the late veneer”, *Nature*, 520 (2015), 534–537. doi:10.1038/nature14360.
- [35] M. Touboul, I. S. Puchtel and R. J. Walker, “Tungsten isotopic evidence for disproportional late accretion to the Earth and Moon”, *Nature*, 520 (2015), 530–533. doi:10.1038/nature14355.
- [36] E. D. Young, I. E. Kohl, P. H. Warren, D. C. Rubie, S. A. Jacobson and A. Morbidelli, “Oxygen isotopic evidence for vigorous mixing during the Moon-forming giant impact”, *Science*, 351 (2016), 493–496. doi:10.1126/science.aad0525.
- [37] J. Zhang, N. Dauphas, A. M. Davis, I. Leya and A. Fedkin, “The proto-Earth as a significant source of lunar material”, *Nature Geoscience*, 5 (2012), 251–255. doi:10.1038/ngeo1429.

- [38] A. E. Ringwood, T. Kato, W. Hibberson and N. Ware, “High-pressure geochemistry of Cr, V and Mn and implications for the origin of the Moon”, *Nature*, 347 (1990), 174–176. doi:10.1038/347174a0.
- [39] M. J. Drake, H. E. Newsom and C. J. Capobianco, “V, Cr, and Mn in the Earth, Moon, EPB, and SPB and the origin of the Moon: experimental studies”, *Geochimica et Cosmochimica Acta*, 53 (1989), 2101–2111.
- [40] C. K. Gessmann and D. C. Rubie, “The origin of the depletions of V, Cr and Mn in the mantles of the Earth and Moon”, *Earth and Planetary Science Letters*, 184 (2000), 95–107.
- [41] H. E. Newsom, “Constraints on the origin of the Moon from the abundance of molybdenum and other siderophile elements”, in W. K. Hartmann, R. J. Phillips and G. J. Taylor (eds.), *Origin of the Moon*, Lunar and Planetary Institute, Houston (1986), 203–229.
- [42] R. M. Canup, K. Righter, N. Dauphas et al., “Origin of the Moon”, *Reviews in Mineralogy and Geochemistry*, 89 (2023), 53–102. arXiv:2103.02045.
- [43] L. Spitzer, “Disruption of galactic clusters”, *The Astrophysical Journal*, 127 (1958), 17–27.
- [44] J. Binney and S. Tremaine, *Galactic Dynamics*, 2nd edition, Princeton University Press, Princeton (2008).
- [45] A. Toomre and J. Toomre, “Galactic bridges and tails”, *The Astrophysical Journal*, 178 (1972), 623–666.
- [46] I. Mosqueira and P. R. Estrada, “Formation of the regular satellites of giant planets in an extended gaseous nebula I: Subnebula model and accretion of satellites”, *Icarus*, 163 (2003), 198–231. doi:10.1016/S0019-1035(03)00076-9.
- [47] A. Crida and S. Charnoz, “Formation of regular satellites from ancient massive rings in the Solar System”, *Science*, 338 (2012), 1196–1199. doi:10.1126/science.1226477.
- [48] C. B. Agnor and D. P. Hamilton, “Neptune’s capture of its moon Triton in a binary–planet gravitational encounter”, *Nature*, 441 (2006), 192–194. doi:10.1038/nature04792.
- [49] D. Banfield and N. Murray, “A dynamical history of the inner Neptunian satellites”, *Icarus*, 99 (1992), 390–401. doi:10.1016/0019-1035(92)90155-Z.
- [50] M. R. Showalter, I. de Pater, J. J. Lissauer and R. S. French, “The seventh inner moon of Neptune”, *Nature*, 566 (2019), 350–353. doi:10.1038/s41586-019-0909-9.
- [51] J. Szulágyi, M. Cilibasi and L. Mayer, “In situ formation of icy moons of Uranus and Neptune”, *The Astrophysical Journal Letters*, 868 (2018), L13. doi:10.3847/2041-8213/aabcd6.
- [52] M. R. Davis, M. Belyakov, I. Wong, Z. Milby and M. E. Brown, “Neptune’s Inner Moons and Rings Are Exposed Icy Body Interiors”, arXiv:2607.27481 (2026).
- [53] K. Tsiganis, R. Gomes, A. Morbidelli and H. F. Levison, “Origin of the orbital architecture of the giant planets of the Solar System”, *Nature*, 435 (2005), 459–461. doi:10.1038/nature03539.
- [54] R. Gomes, H. F. Levison, K. Tsiganis and A. Morbidelli, “Origin of the cataclysmic Late Heavy Bombardment period of the terrestrial planets”, *Nature*, 435 (2005), 466–469. doi:10.1038/nature03676.
- [55] D. Nesvorný, “Young Solar System’s fifth giant planet?”, *The Astrophysical Journal Letters*, 742 (2011), L22. doi:10.1088/2041-8205/742/2/L22.
- [56] D. Nesvorný and A. Morbidelli, “Statistical study of the early Solar System’s instability with four, five, and six giant planets”, *The Astronomical Journal*, 144 (2012), 117. doi:10.1088/0004-6256/144/4/117.
- [57] B. Liu, S. N. Raymond and S. A. Jacobson, “Early Solar System instability triggered by dispersal of the gaseous disk”, *Nature*, 604 (2022), 643–646. doi:10.1038/s41586-022-04535-1.
- [58] M. S. Clement, N. A. Kaib, S. N. Raymond and K. J. Walsh, “Mars’ growth stunted by an early giant planet instability”, *Icarus*, 311 (2018), 340–356. doi:10.1016/j.icarus.2018.04.008.